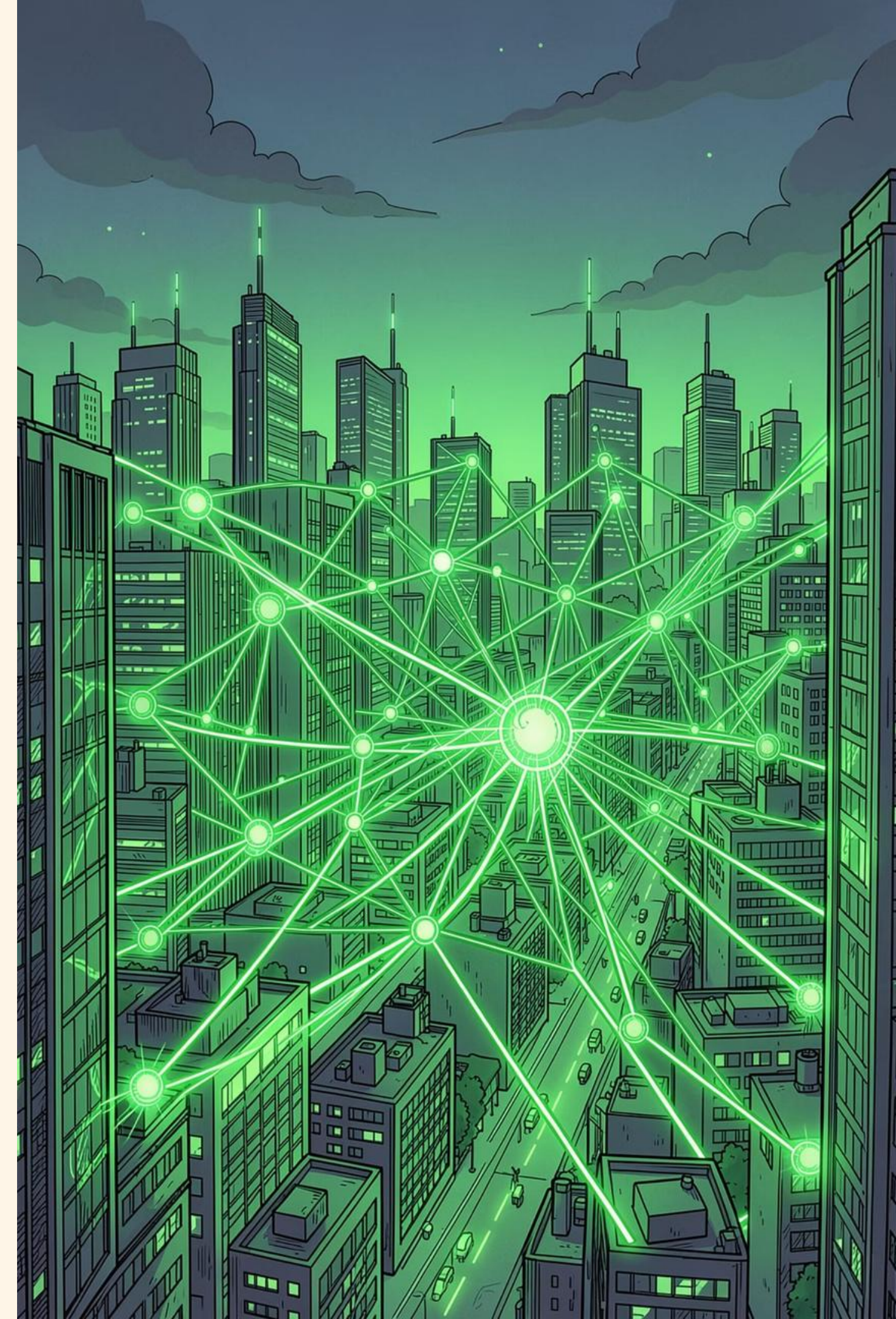
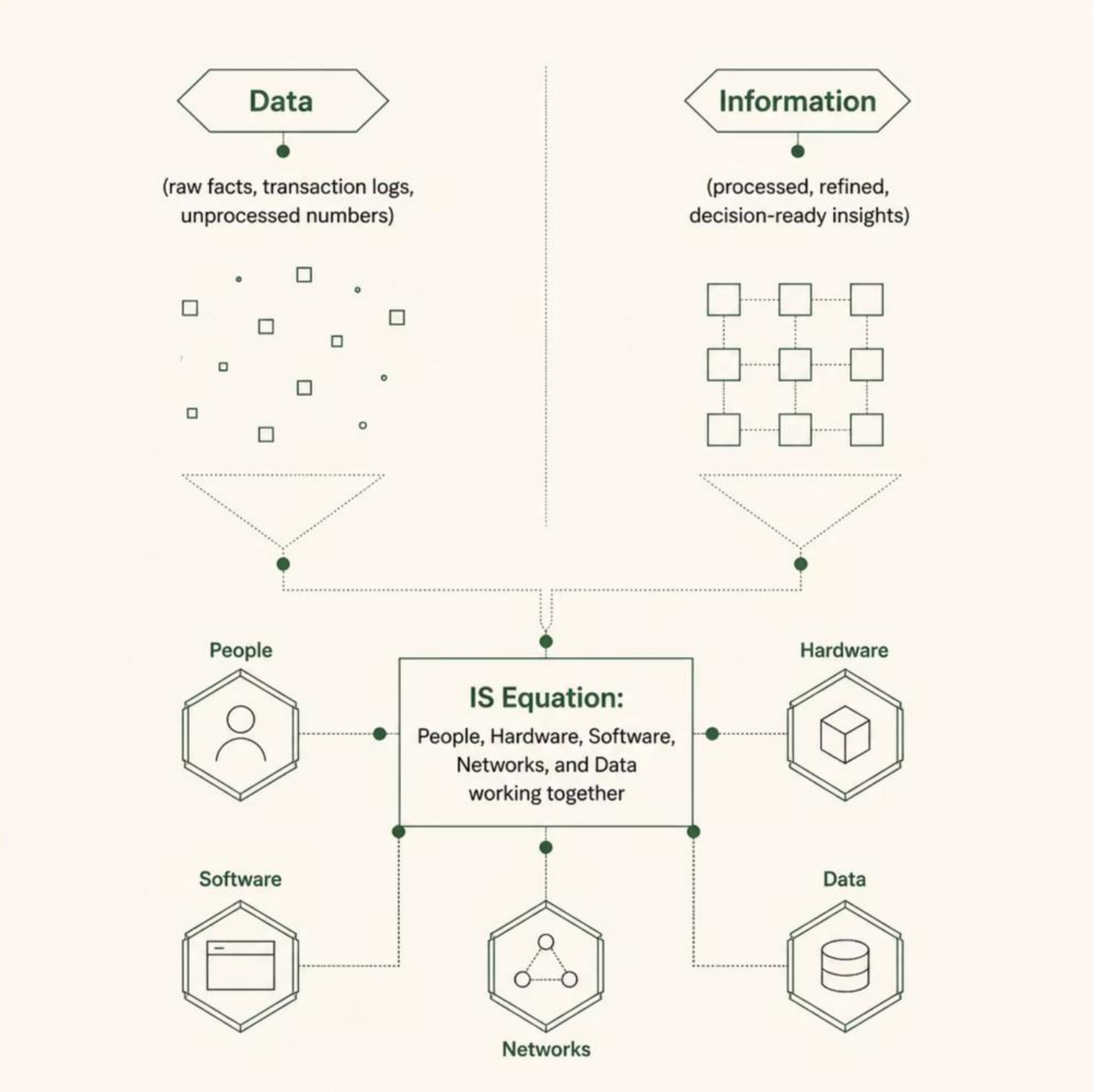


Introduction to IT & Business Systems

The digital foundation shaping every modern organisation



Data vs. Information



The IS Equation

Five components working in unison:

People	Hardware
Software	Networks
Data	

From Tabulators to Networks

1

1890s

Hollerith's census tabulator — first large-scale system

2

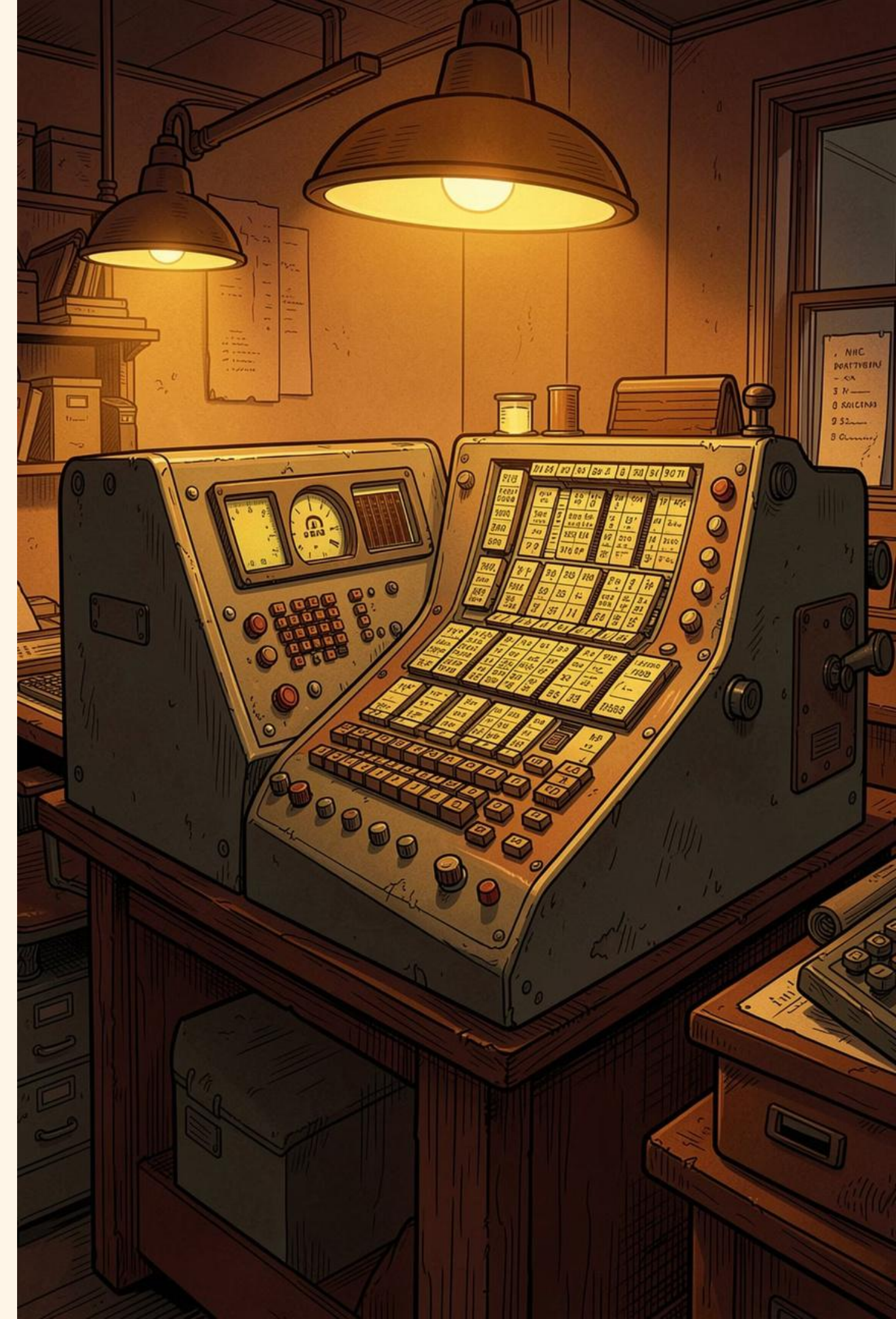
1950s

Data processing & basic record-keeping

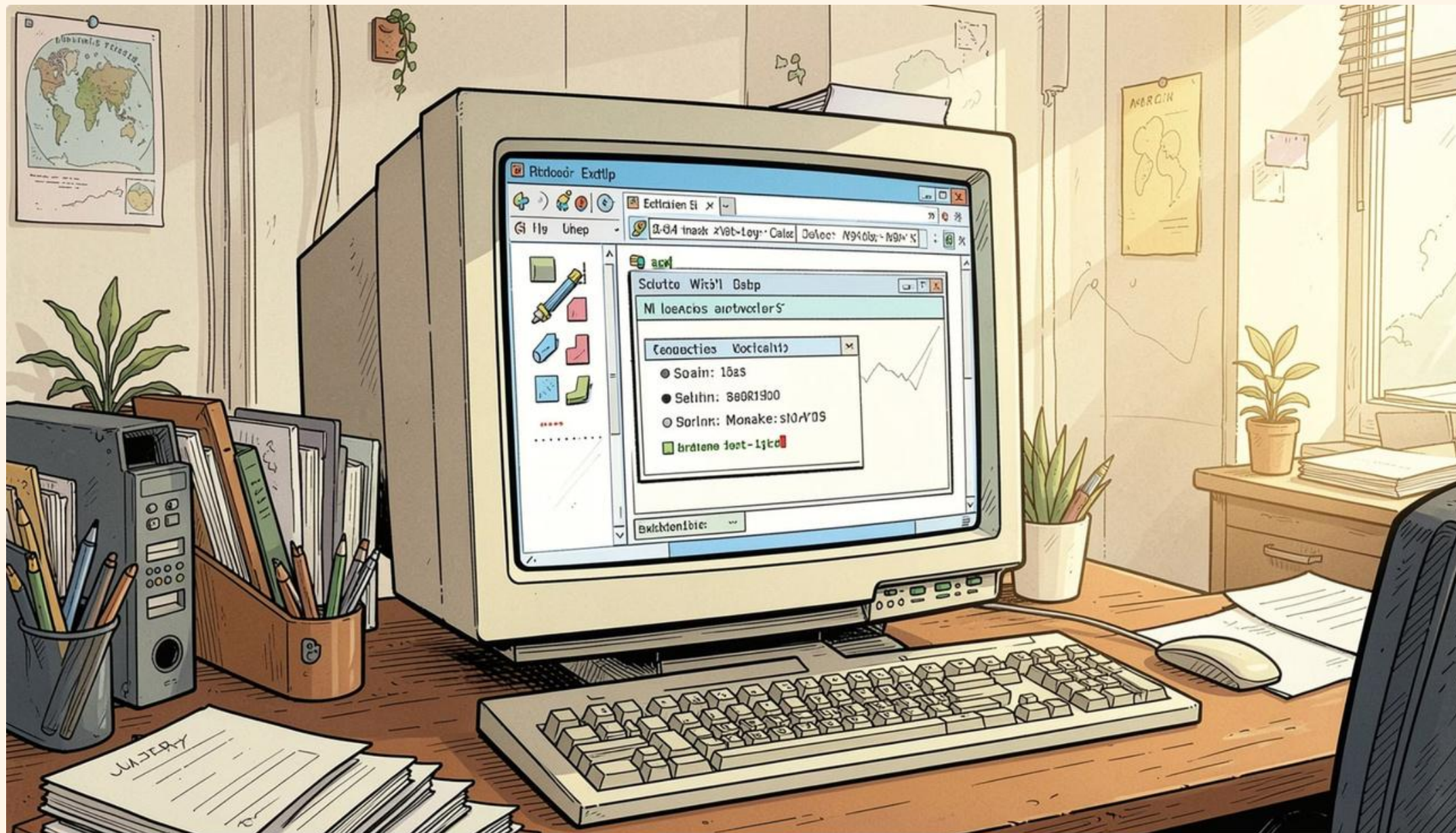
3

1960s–70s

Management reporting & decision support systems



The Web Revolution: 1990–2005



1990 — The Web is Born

Tim Berners-Lee introduces the World Wide Web at CERN

2005 — Web 2.0 Arrives

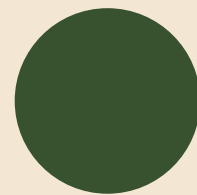
Collective intelligence, social collaboration, user-generated content

The Shift

Closed systems → interconnected global platforms

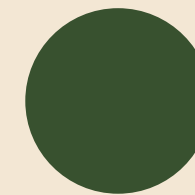


Why Organisations Invest in IT



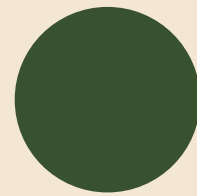
Operational Excellence

Efficiency & productivity gains



Competitive Advantage

Outperform rivals through technology



Strategic Necessity

Survival in a digital-first market

Transforming Enterprise Architecture



Flattened Structures

Hierarchical → decentralised organisations

Location Independence

Digital networks enable remote & global teams

The Digital Firm

Core processes integrated via real-time data

The Strategic Toolkit



ERP

Enterprise Resource Planning — process automation across functions



Data Warehousing & Mining

Uncovering hidden insights from vast datasets



DSS

Decision Support Systems — modelling "what-if" scenarios

The Modern Risk Landscape

Cybersecurity Threats

Phishing, identity theft, and the rising cost of connectivity

IP Threats

Intellectual property risks in a globalised digital market

Defence Essentials

Firewalls, encryption, and robust security policies



Future Horizons: The Intelligent Web



What's Changing

Systems evolve from reactive to **predictive** — powered by machine-understandable semantics and AI integration.

Web 3.0 enables automation of complex tasks through semantic understanding.

The Competitive Digital Edge

IT is the Engine

No longer a support function —
the primary vehicle for
innovation

Sociotechnical Fit

Balance technology with human
potential for lasting success

Strategic Value

Harness these tools to drive long-term competitive advantage



Data vs. Information

Data

Raw facts & figures — transaction logs, sensor readings, numbers without context.

Information

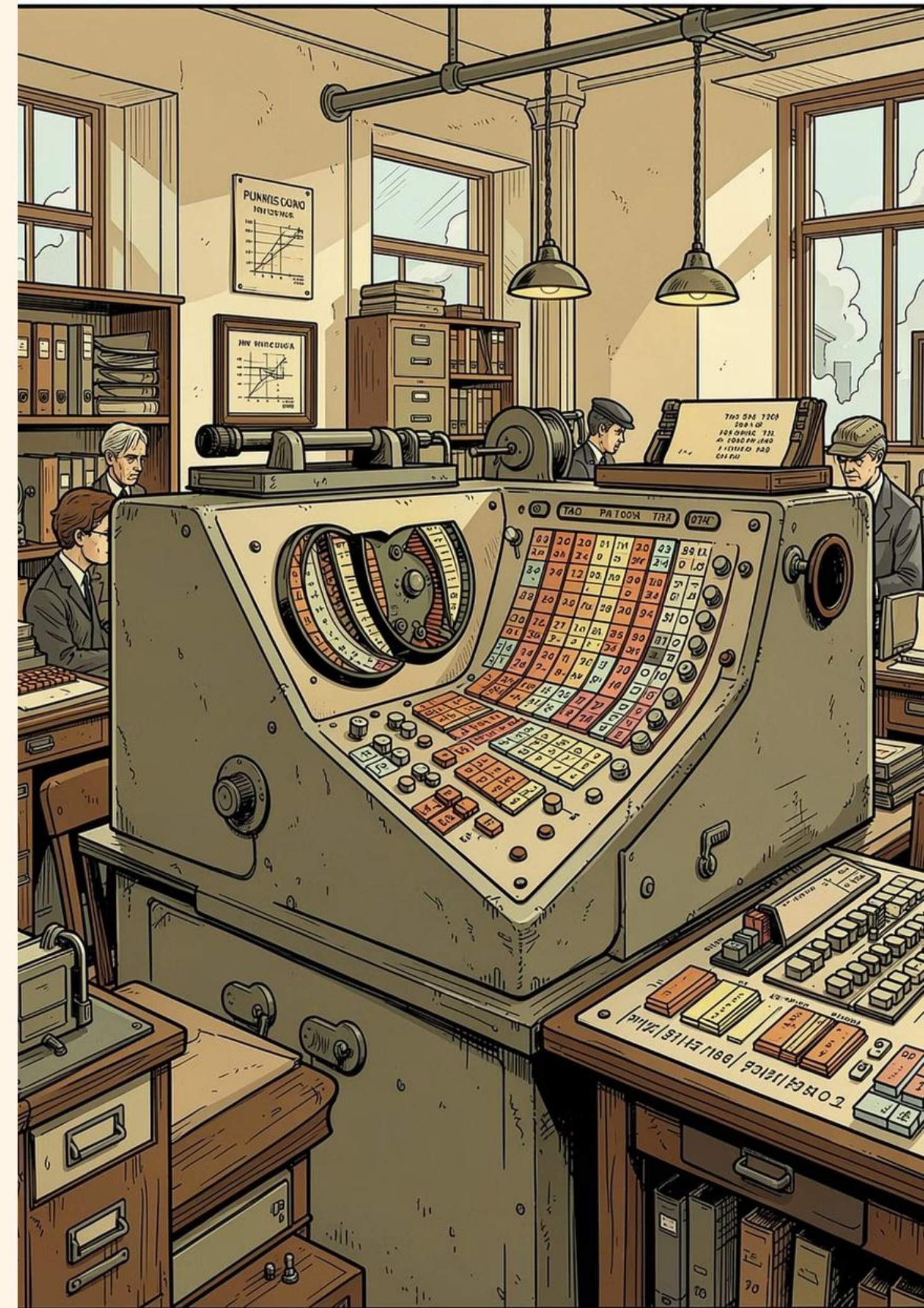
Processed data refined for decision-making — meaningful, actionable, contextual.

The IS Equation

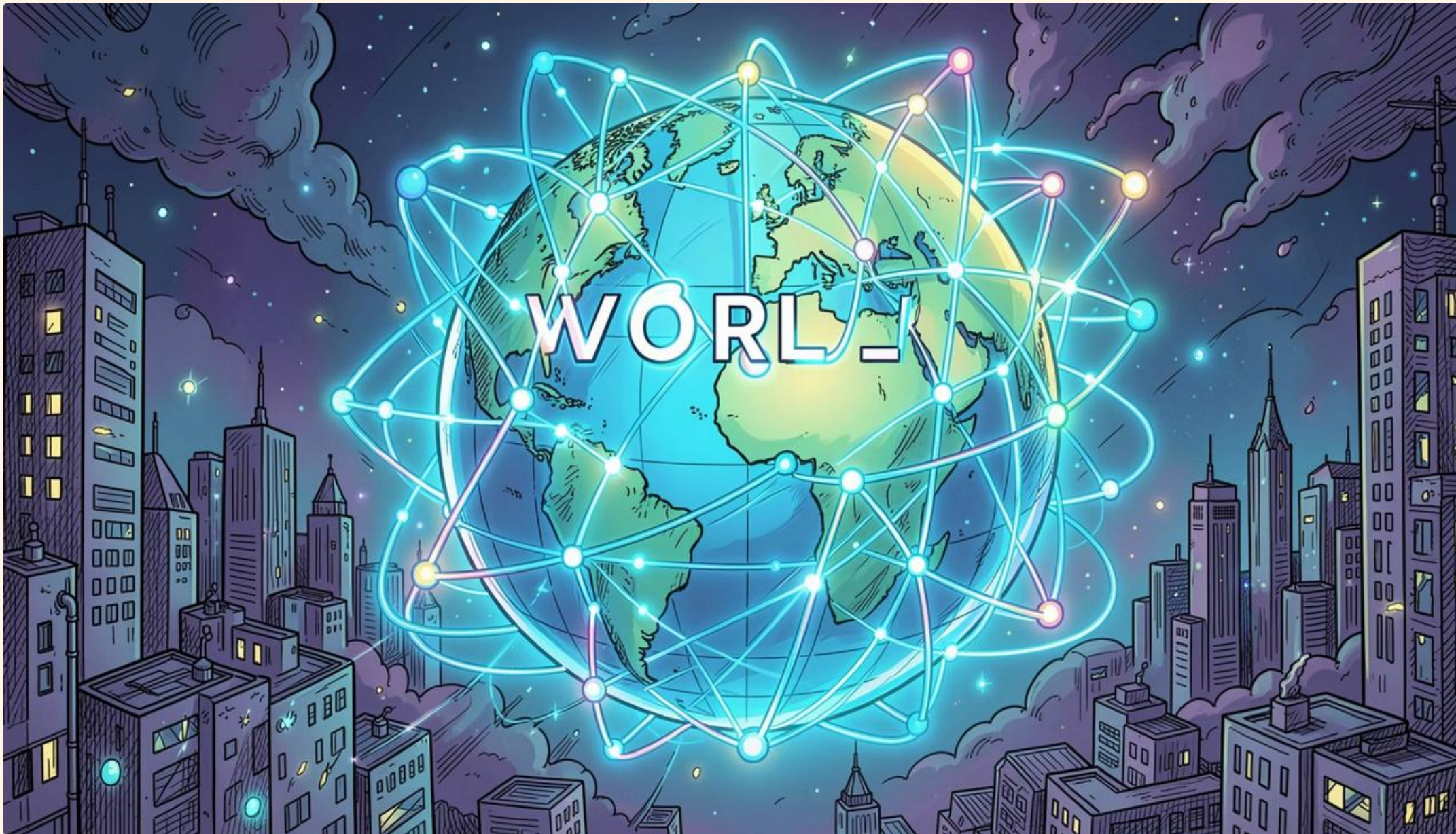
People · Hardware · Software · Networks · Data — working in unison.

From Tabulators to Networks

- 1 — 1890s
Hollerith's census tabulator — first large-scale data system
- 2 — 1950s
Data processing era — basic record-keeping automation
- 3 — 1960s–70s
Management reporting & decision support systems emerge



The Web Revolution: 1990–2005



1990

Tim Berners-Lee launches the World Wide Web at CERN

2005

Web 2.0 arrives — collective intelligence, social collaboration, user-generated content

The Shift

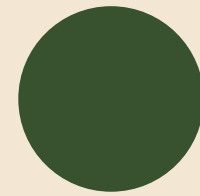
Closed systems → interconnected global platforms

Why Organizations Invest in IT



Operational Excellence

Efficiency & productivity at scale



Competitive Advantage

Technology to outperform rivals

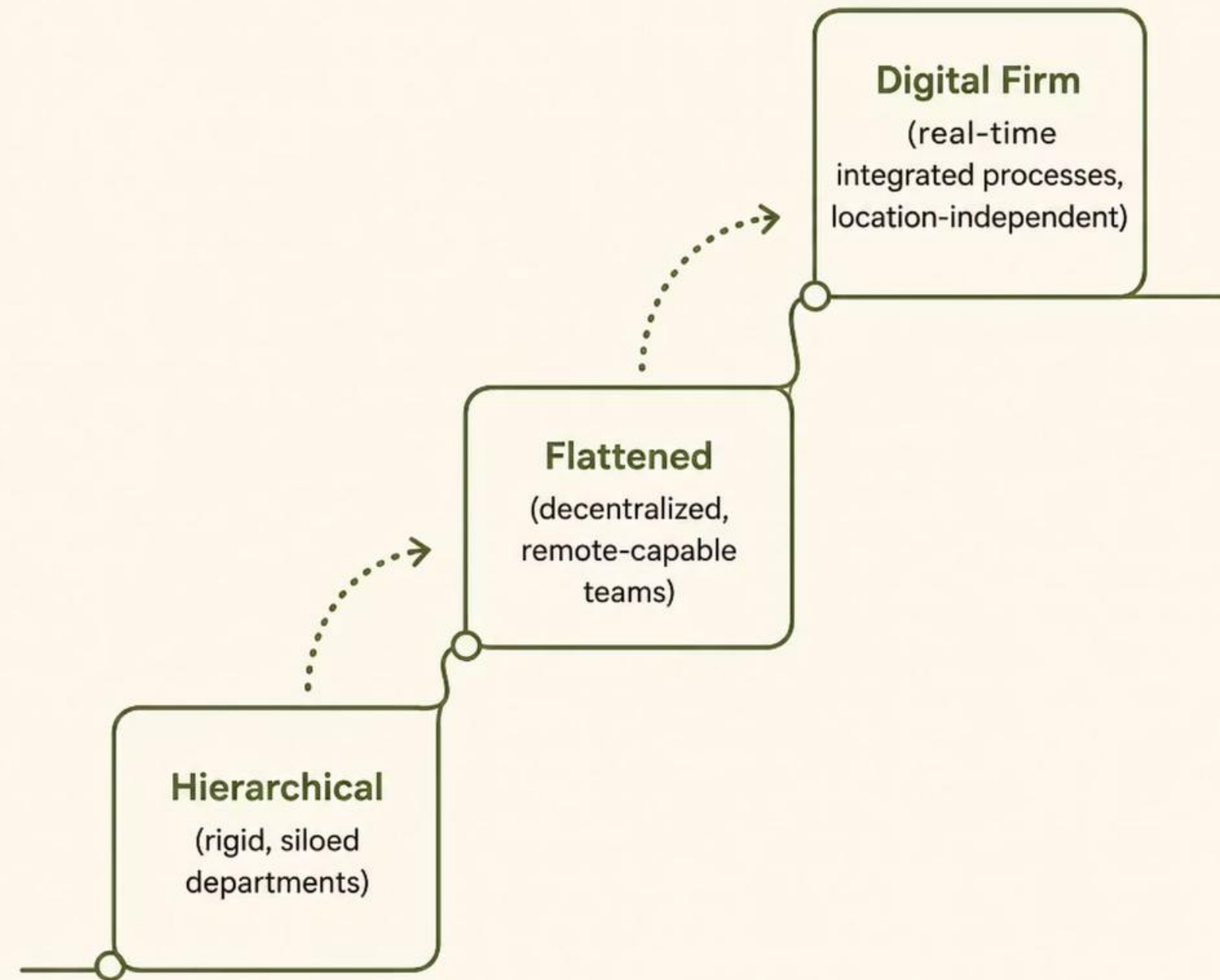


Strategic Necessity

Survival in a digital-first market



Transforming Enterprise Architecture

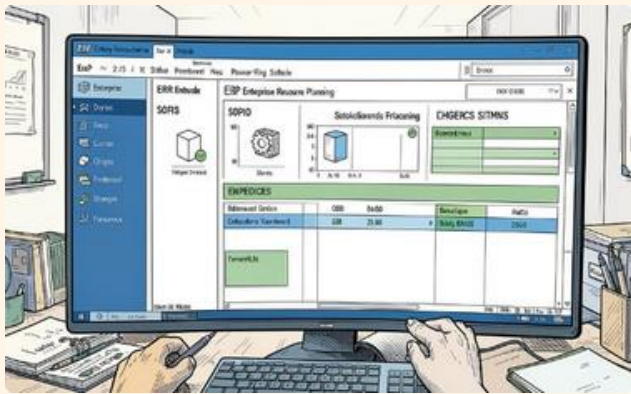


The Digital Firm

Core business processes integrated through real-time data — enabling global teams, remote work, and instant decision-making.

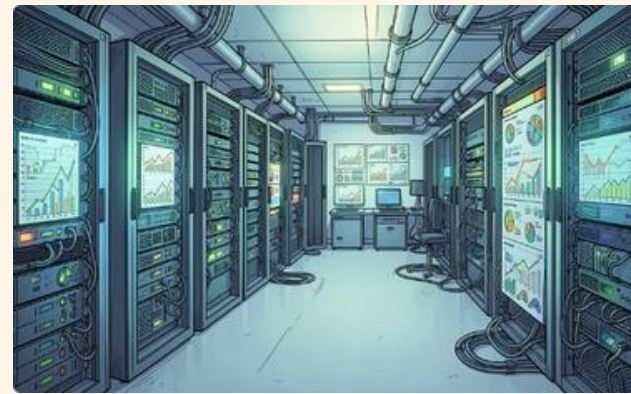
- Hierarchical → flattened structures
- Location independence via digital networks
- Real-time process integration

The Strategic Toolkit



ERP

Enterprise Resource Planning — automates and integrates core business processes



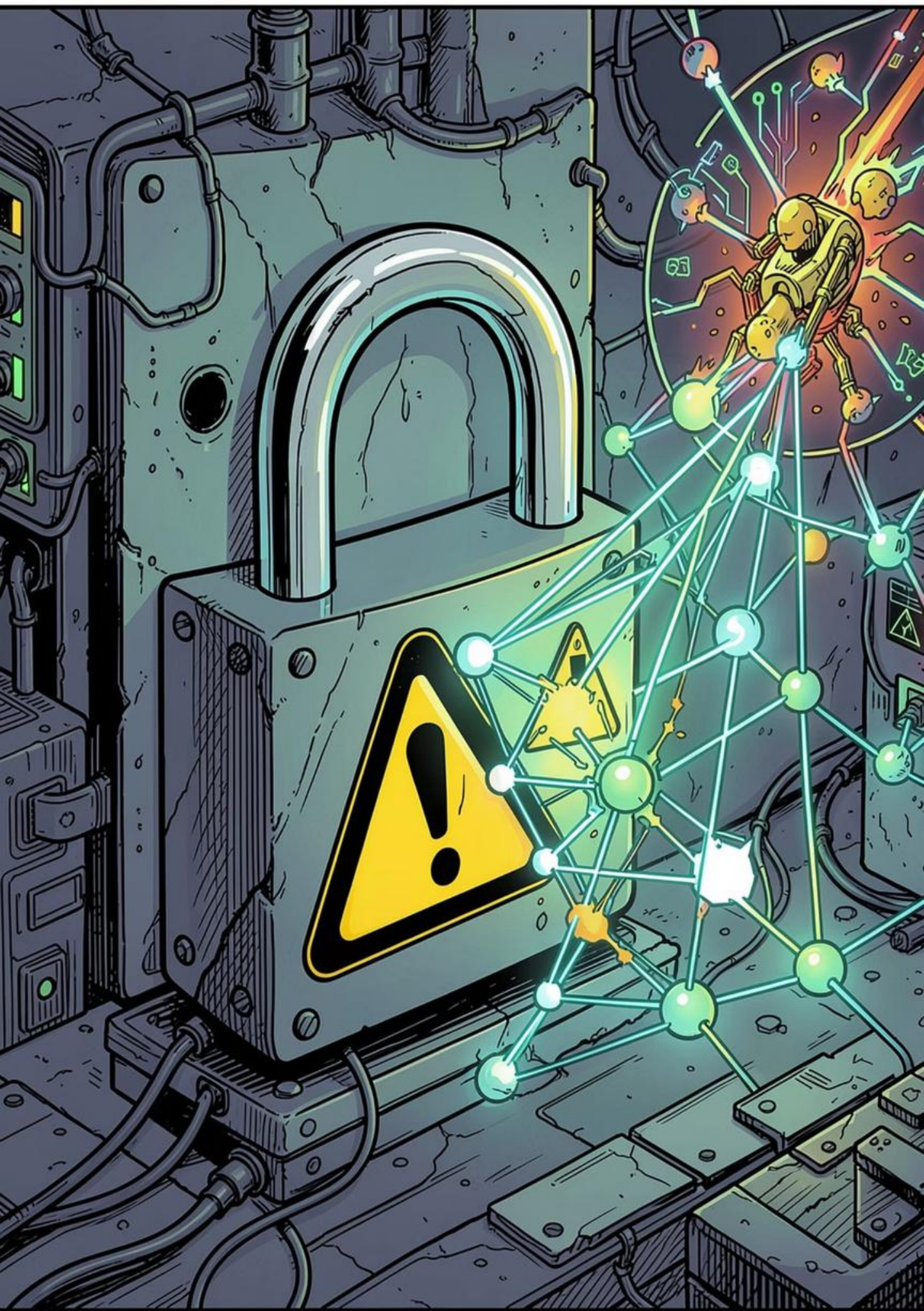
Data Warehousing & Mining

Uncovers hidden patterns and insights from vast datasets



DSS

Decision Support Systems — model "what-if" scenarios for strategic choices



The Modern Risk Landscape



Connectivity Risks

Cybersecurity threats, phishing, identity theft



IP Threats

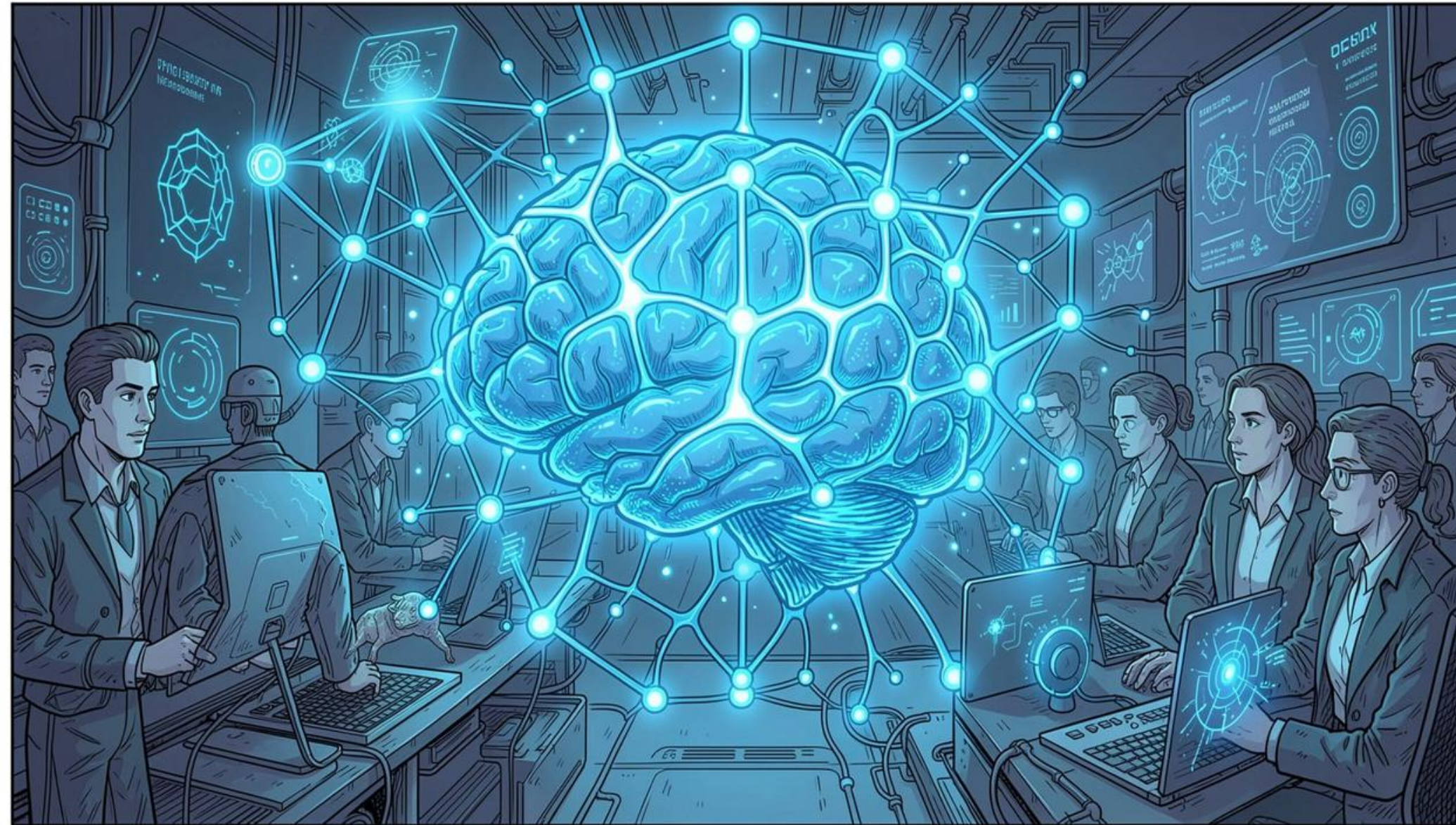
Intellectual property exposure in global markets



Defence Layers

Firewalls, encryption & robust security policies

Future Horizons: The Intelligent Web



Semantic Web (Web 3.0)

Systems that understand the *meaning* of data, not just keywords

Machine-Understandable Semantics

Automation of complex tasks through intelligent data interpretation

Predictive Operations

AI integration moves companies from reactive to proactive strategies

The Competitive Digital Edge

IT = Innovation

No longer a support function —
the primary driver of strategic
value

Sociotechnical Fit

Balance technology with human
potential for lasting success

Act Now

Harness these tools to build long-term competitive advantage



Introduction to the IT Business Analyst Role

Bridge the gap between business needs and technology solutions

CAREER FOUNDATIONS



What Is a Business Analyst?



A BA **identifies business needs** and determines solutions to business problems — often through technology.

Translator

Between stakeholders and developers

Investigator

Uncovers root causes, not just symptoms

Facilitator

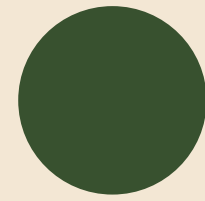
Guides teams toward shared understanding

Core Responsibilities



Requirements Gathering

Interviews, workshops, surveys



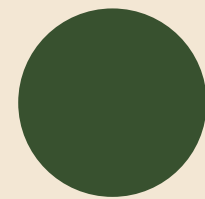
Process Analysis

Map and improve workflows



Solution Design

Define functional specifications



Validation & Testing

Ensure solutions meet requirements



Key Skill #1: Communication

Written

Reports, specs, user stories

Verbal

Presentations, facilitation

Visual

Diagrams, wireframes, models

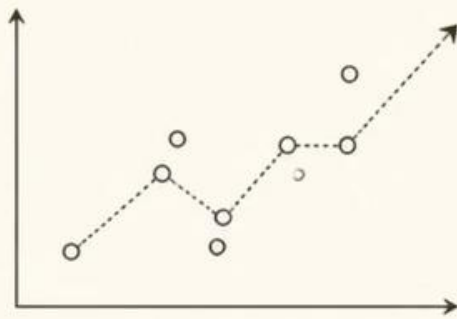
Key Skill #2: Analytical Thinking



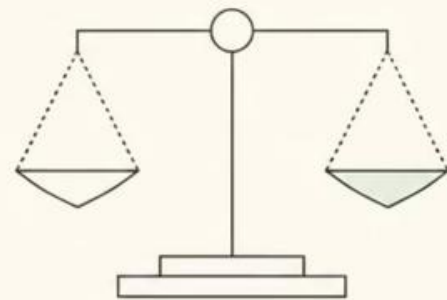
Data Interpretation
(reading charts and metrics)



Root Cause Analysis
(asking why repeatedly)



Pattern Recognition
(spotting trends in data)



Critical Evaluation
(assessing solutions objectively)

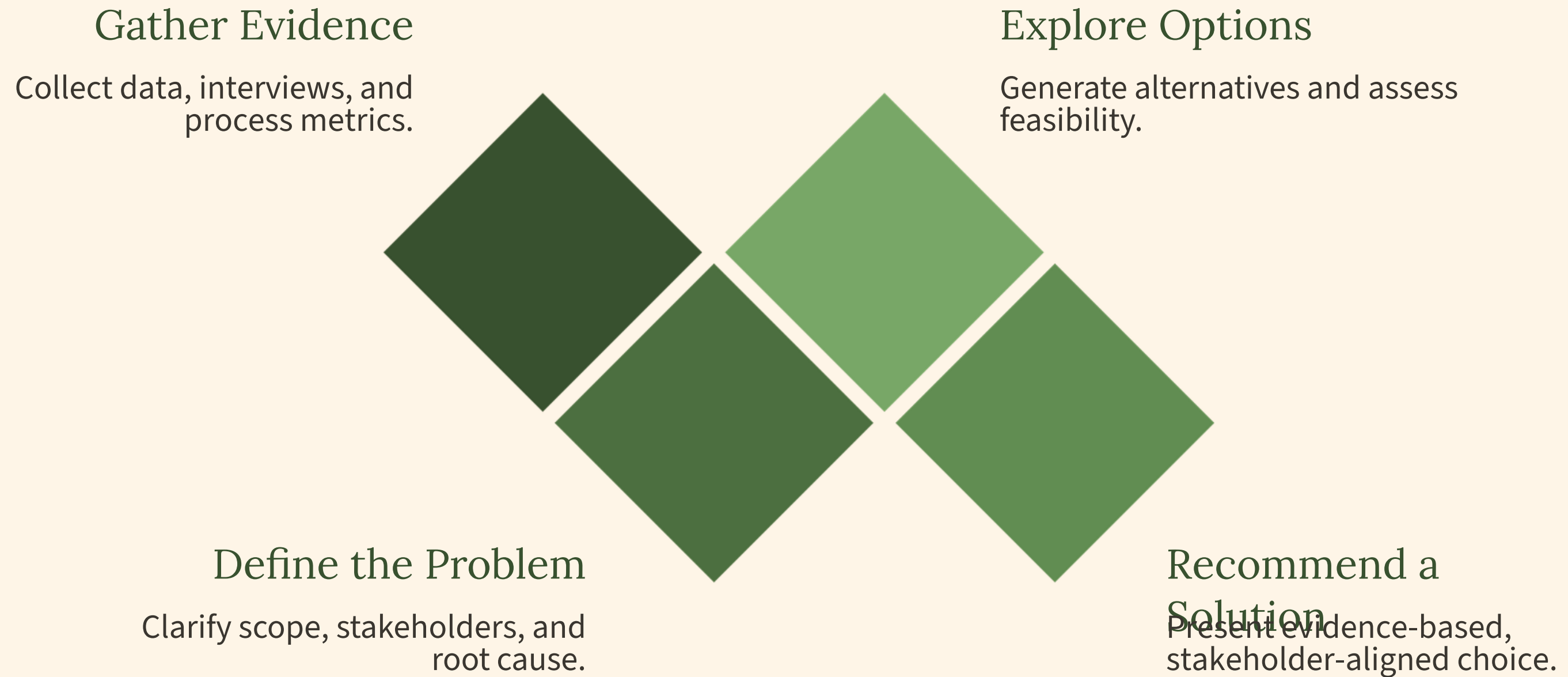
Think Clearly, Decide Confidently

BAs must cut through complexity to find what truly matters — then communicate it simply.



Strong analysts ask **"Why?"** at least five times before proposing a solution.

Key Skill #3: Problem-Solving



Effective BAs don't just identify problems — they guide stakeholders toward **actionable, evidence-based solutions**.

Business Analysis Methodologies

Agile

Iterative delivery, frequent feedback, adaptable scope

- Sprints & stand-ups
- User stories & backlogs
- Continuous stakeholder input

Waterfall

Sequential phases, fixed requirements, structured documentation

- Requirements → Design → Build → Test
- Clear milestones & sign-offs
- Best for stable, well-defined projects



Agile vs. Waterfall: When to Use Which



Choose Agile When...

Requirements evolve, speed matters, and collaboration is high



Choose Waterfall When...

Scope is fixed, compliance is critical, and documentation is required

Stakeholder Management



BAs work with **diverse stakeholders** — from end users to C-suite executives. Managing expectations is key.

Identify

Who has influence? Who is impacted?

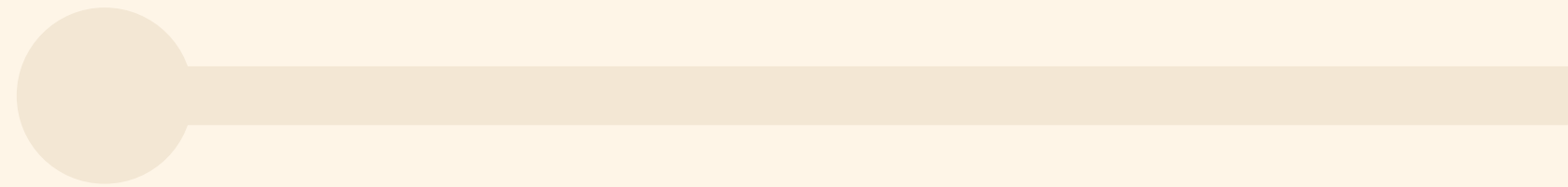
Engage

Regular updates, active listening

Align

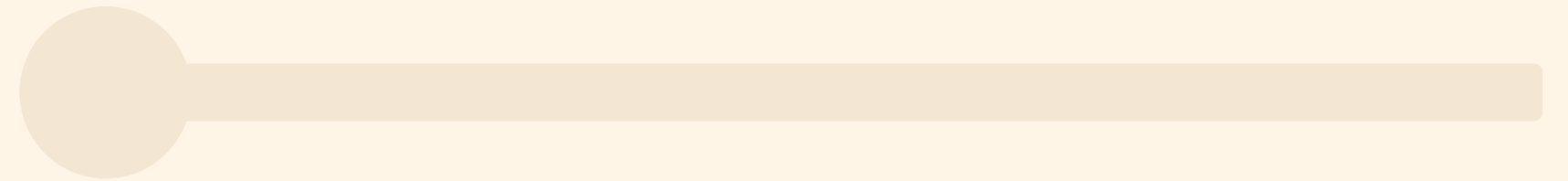
Resolve conflicts, build consensus

Your BA Journey Starts Here



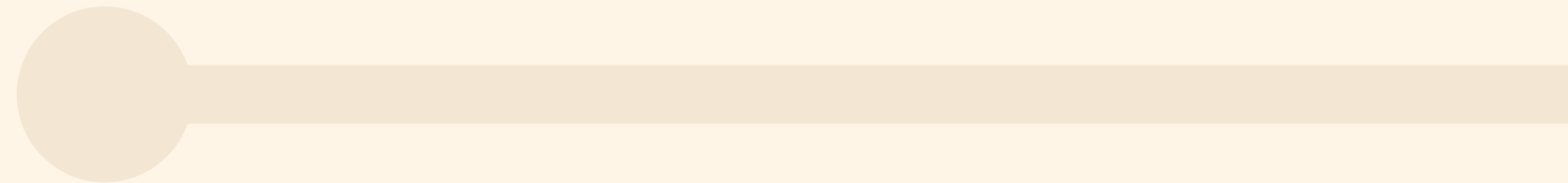
Build Your Foundation

Learn BA frameworks, tools, and techniques



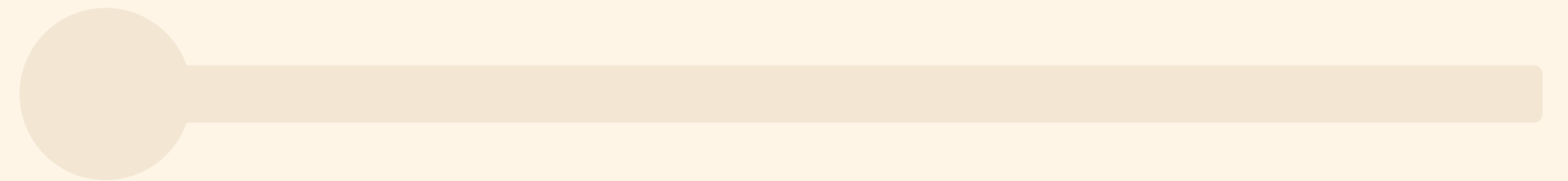
Practice the Skills

Communication, analysis, problem-solving



Earn Recognition

Certifications like ECBA or CBAP



Launch Your Career

Entry-level BA, junior analyst, product owner

An abstract digital illustration on the left side of the slide. It features a dark background with glowing green lines and nodes, resembling a network or data flow. Binary code (0s and 1s) is visible, along with stylized representations of data streams and connections.

Data vs. Information

Understanding the foundation of IT — from raw facts to meaningful insight

IT FUNDAMENTALS · 2 UNITS

What Is Raw Data?

Unprocessed

Collected but not yet organised

Discrete Facts

Individual values with no meaning on their own

Examples

42, "Red", 03/07/2024, 1500



What Is Information?



$\text{Data} + \text{Context} = \text{Information}$

Information is data that has been **processed, organised, and structured** to carry meaning.



"1500" is data. "1500 units sold in March" is information.

Data vs. Information — Side by Side

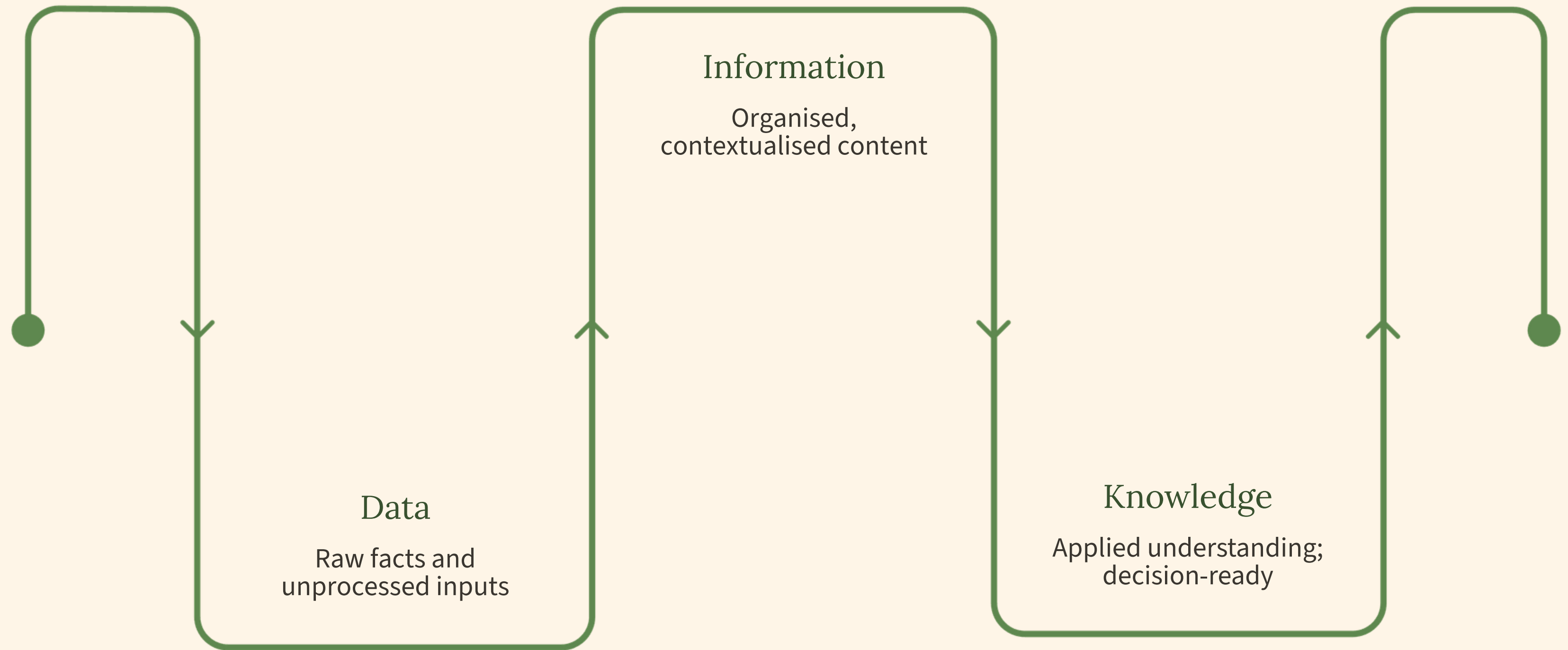
Data

- Raw and unorganised
- No inherent meaning
- Example: 37.5, "Fever"

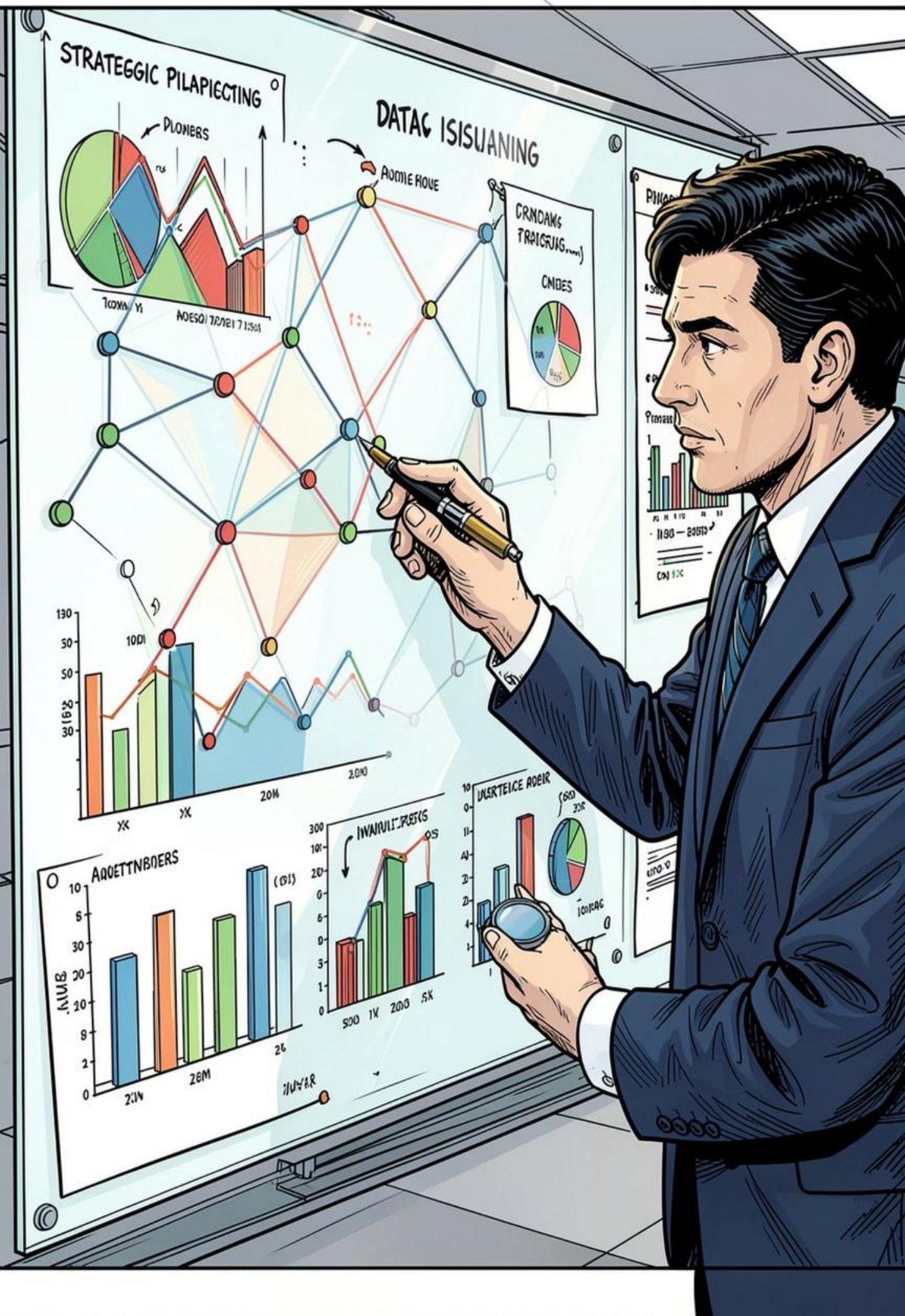
Information

- Processed and structured
- Provides context and meaning
- Example: "Patient has a fever of 37.5°C"

The Transformation Process



Each stage adds value — raw data becomes actionable insight through processing and interpretation.



From Information to Knowledge

Recognise Patterns

Spot trends within information

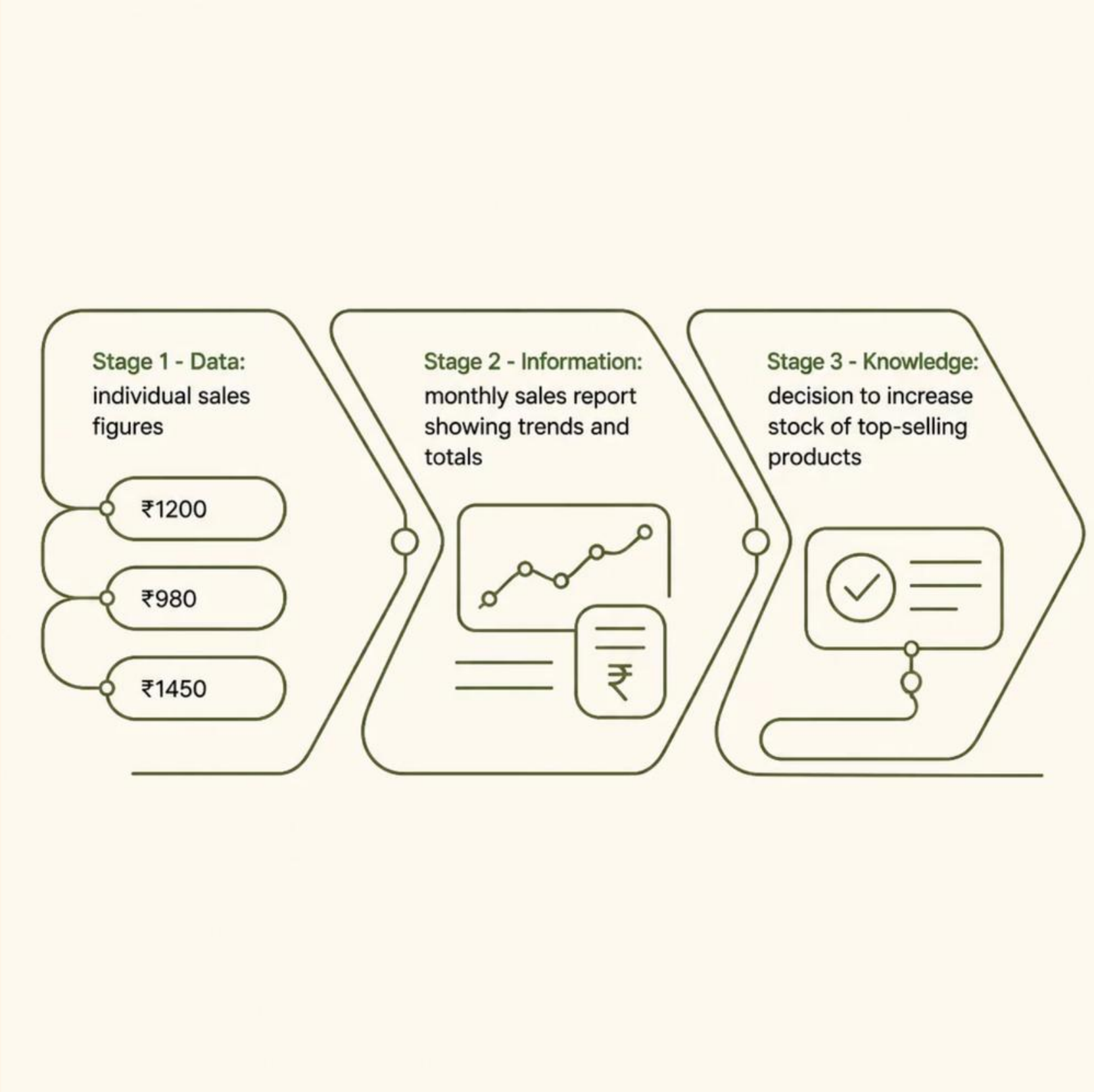
Apply Experience

Use context and judgement

Enable Decisions

Turn insight into action

Real-World Example



From Numbers to Decisions

A retail store collects daily sales figures (data), compiles a monthly report (information), and uses it to decide which products to reorder (knowledge).



Why Information Drives Decisions

Accuracy

Reduces guesswork and errors

Speed

Faster, confident responses

Strategy

Supports long-term planning

The Value of Good Information

Timely

Available when needed

Accurate

Free from errors

Relevant

Directly applicable to the decision

Complete

Nothing critical is missing

Key Takeaways

01

Data is raw

Unprocessed facts with no meaning alone

03

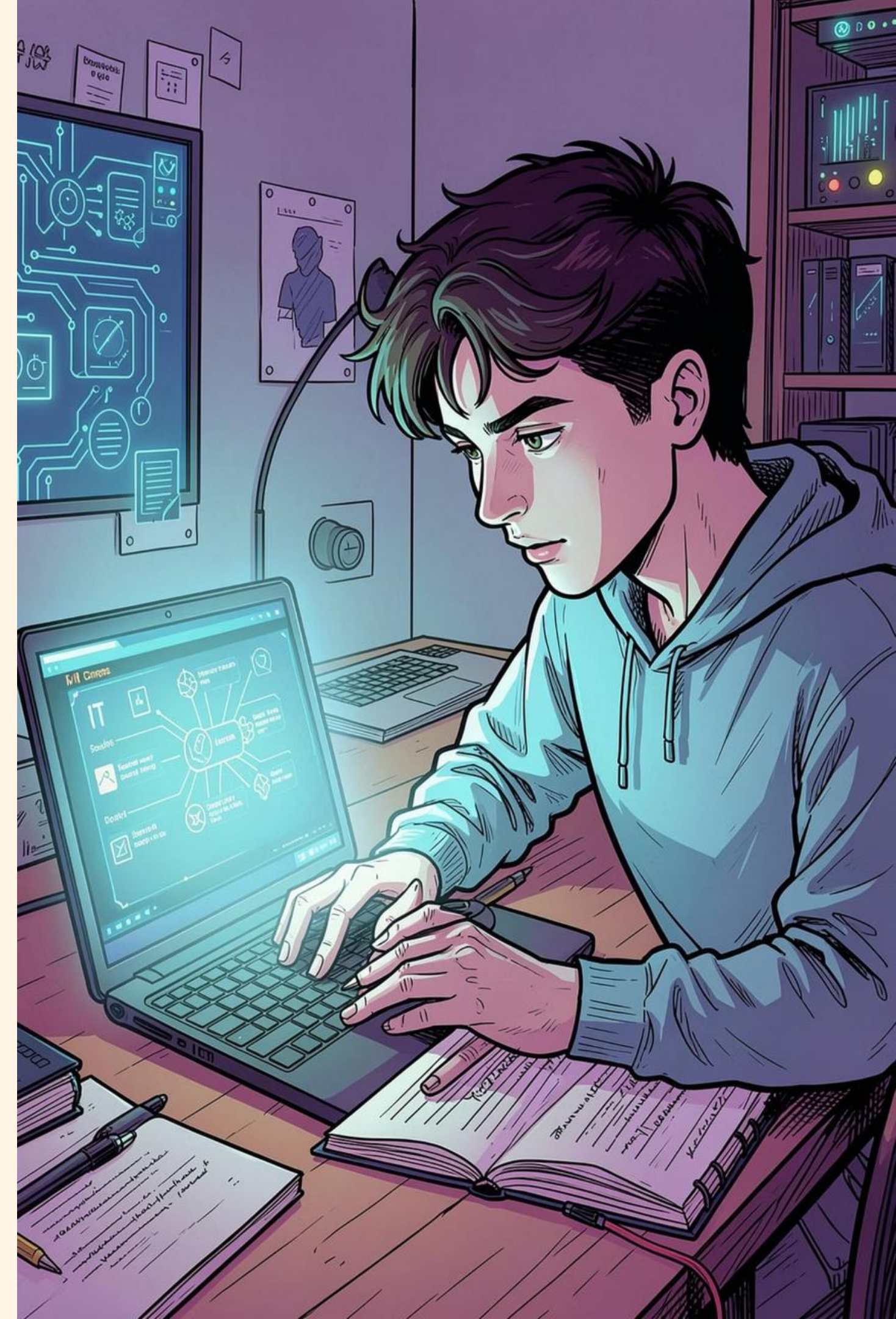
Knowledge drives action

Insight applied to real decisions

02

Information adds context

Organised data becomes meaningful





Data-Driven Decision Making in IT

Transforming how IT teams operate — from intuition to insight.

What Is Data-Driven Decision Making?



Using **collected data, analysis, and evidence** to guide strategic and operational choices — replacing guesswork with measurable insight.

Evidence over instinct

Metrics over assumptions

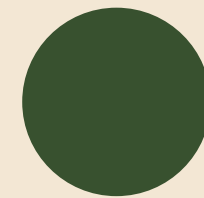
Outcomes over opinions

Core Principles



Define the Question First

Start with a clear problem statement



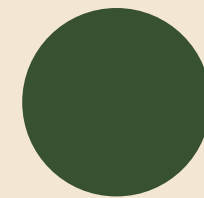
Collect Relevant Data

Quality and relevance over volume



Analyse Objectively

Remove bias from interpretation



Iterate and Improve

Decisions inform future data needs

BENEFITS

Why It Matters for IT Teams

3×

Faster Resolution

Incident response with data-backed
triage

40%

Cost Reduction

Optimised resource allocation

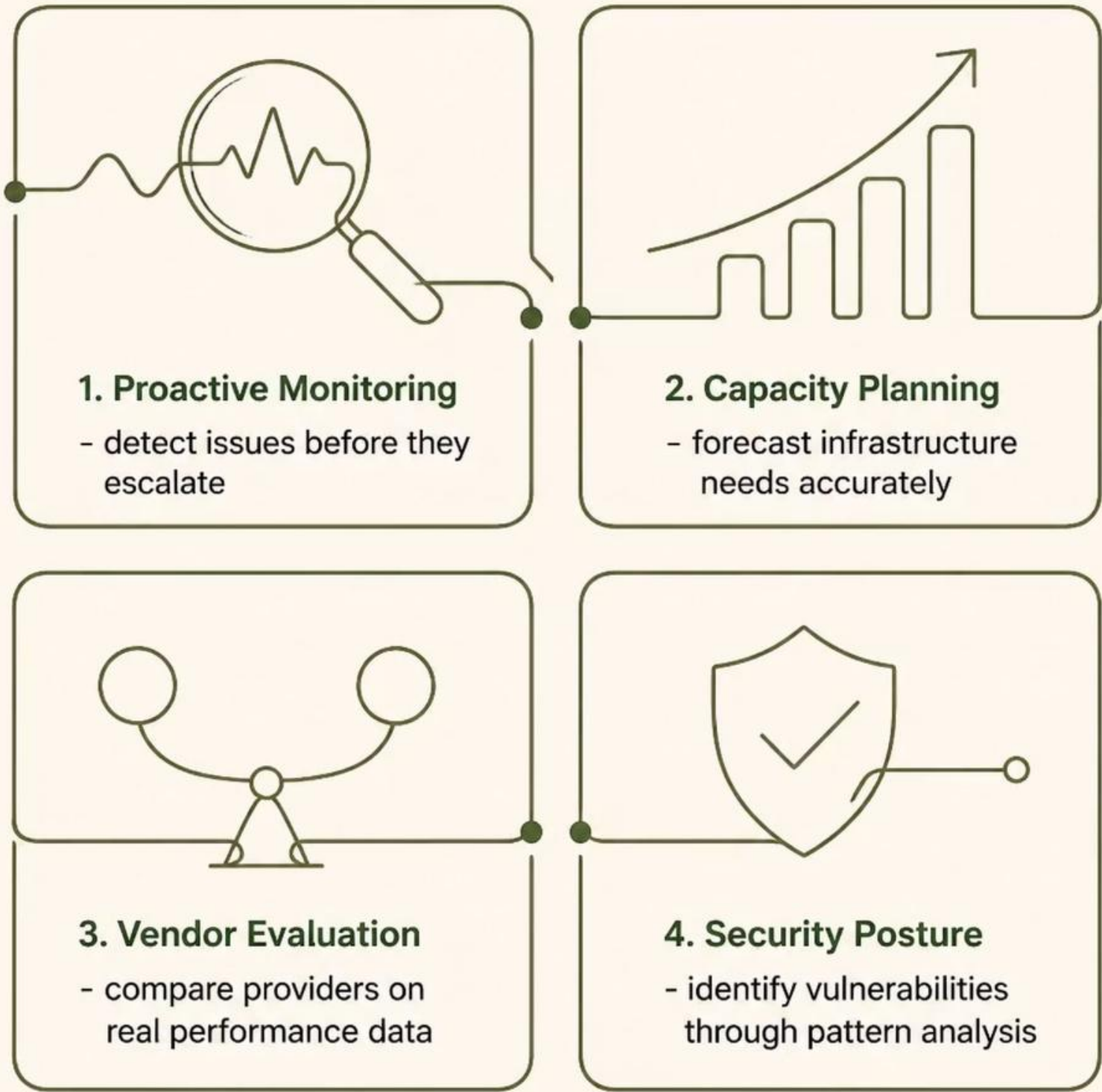
90%

Stakeholder Trust

Evidence-backed proposals win
approval



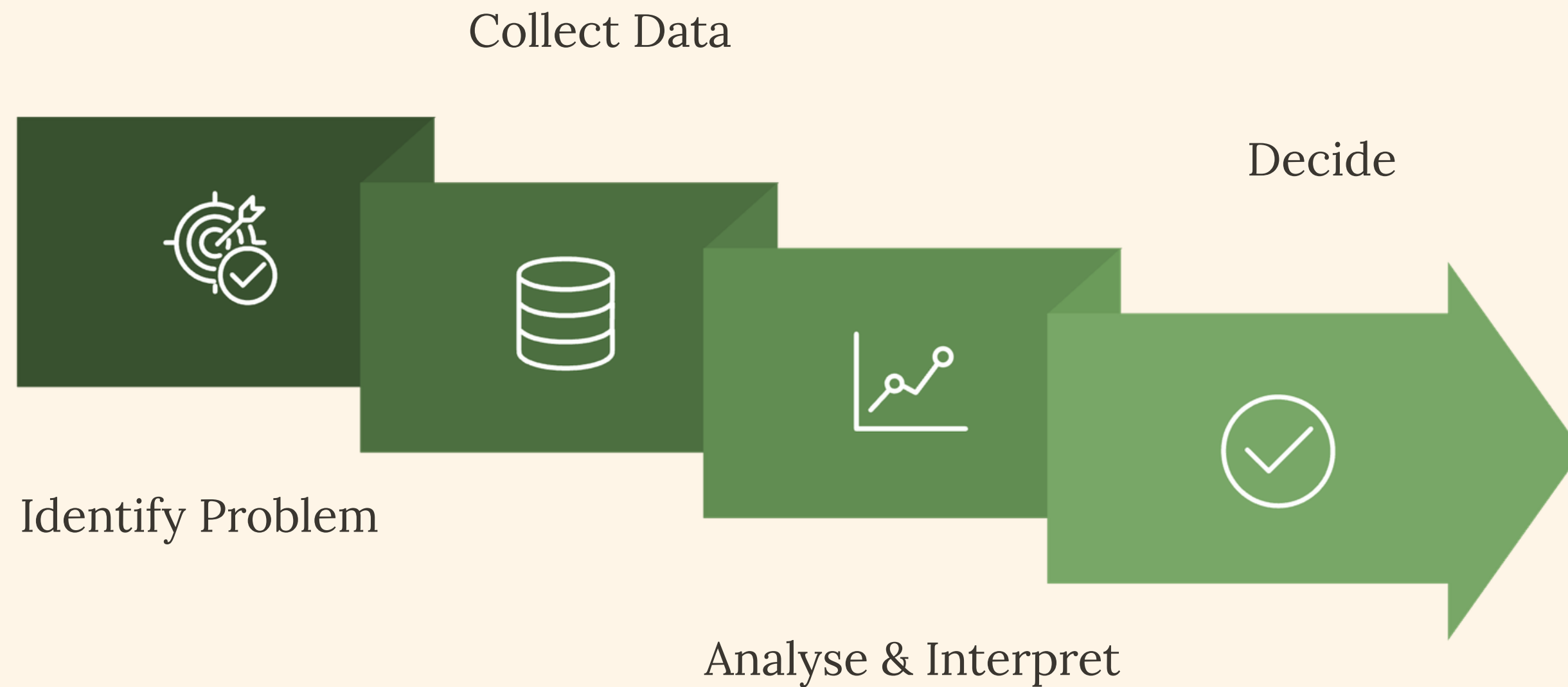
Operational Gains



From Reactive to Proactive

Data shifts IT from firefighting to **strategic planning** — anticipating problems before they impact users.

The Decision-Making Process



Collecting & Analysing Data

Sources of IT Data

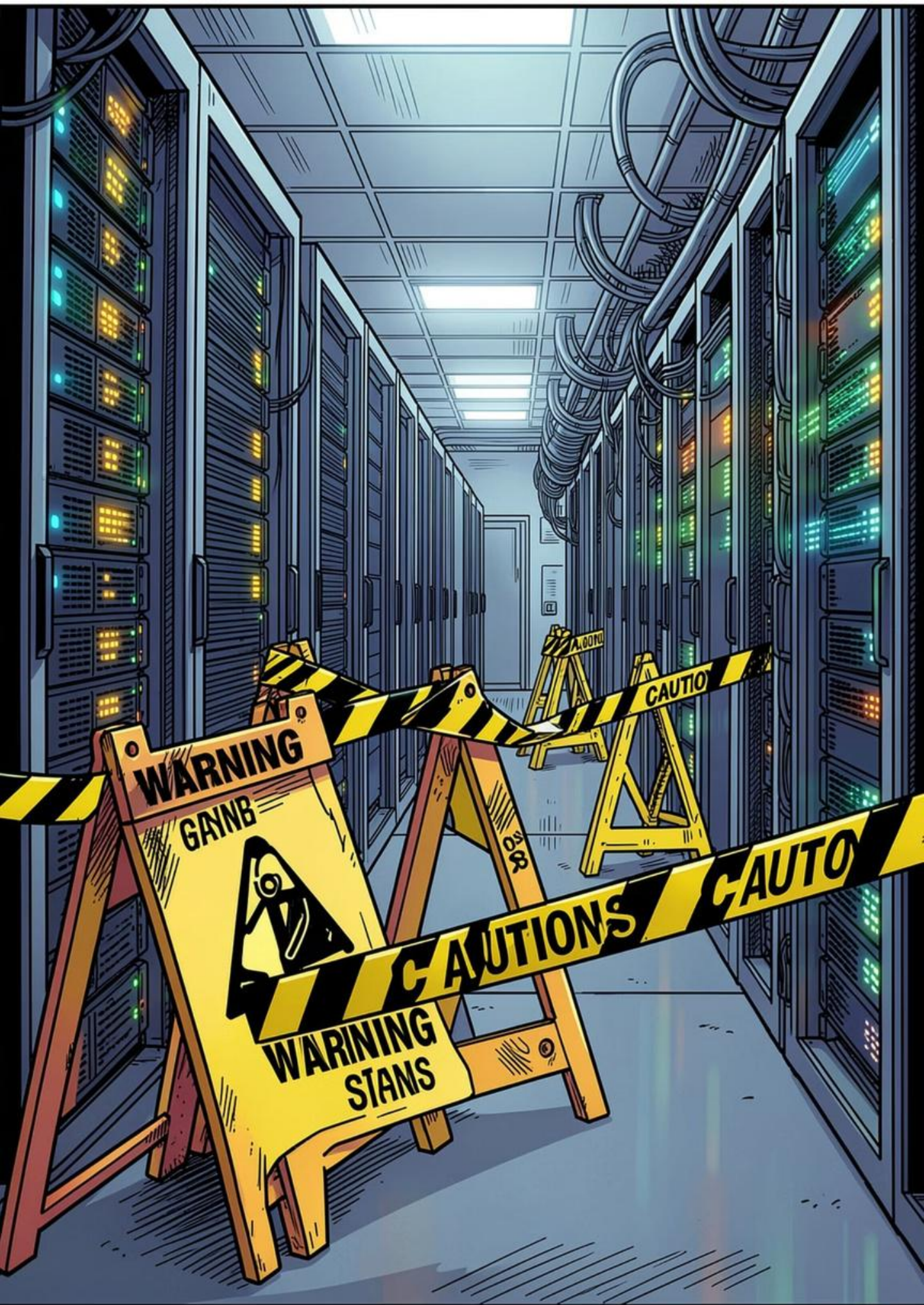
System logs & monitoring tools

Helpdesk & ticketing systems

Network performance metrics

User feedback & surveys





CHALLENGES

Common Pitfalls to Avoid

- 1 Poor Data Quality
Inaccurate or incomplete data leads to flawed conclusions
- 2 Analysis Paralysis
Over-analysing delays action — know when to decide
- 3 Confirmation Bias
Selecting data that supports pre-existing beliefs
- 4 Lack of Context
Numbers without business context mislead decisions

OVERCOMING CHALLENGES

Building a Data-Driven Culture



Governance

Define data ownership and standards

Literacy

Train teams to read and question data

Integration

Connect tools for a single source of truth

KEY TAKEAWAYS

Start Your Data Journey Today

Principles

Evidence over instinct

Benefits

Faster, smarter decisions

Process

Define → Collect → Analyse → Act

Watch Out

Quality, bias, and context





The Anatomy of IT Success and Failure

Real-world lessons from projects that succeeded — and those that didn't.

CASE STUDIES

IT STRATEGY



The Hidden Cost of Assumptions

Rich Con Steel

ERP transition failed — lost orders, billing errors, zero training

The Root Cause

Human factors, not technology

The Fix

Phased transitions over abrupt cutovers

Chicago Public Schools: A Systemic Overload



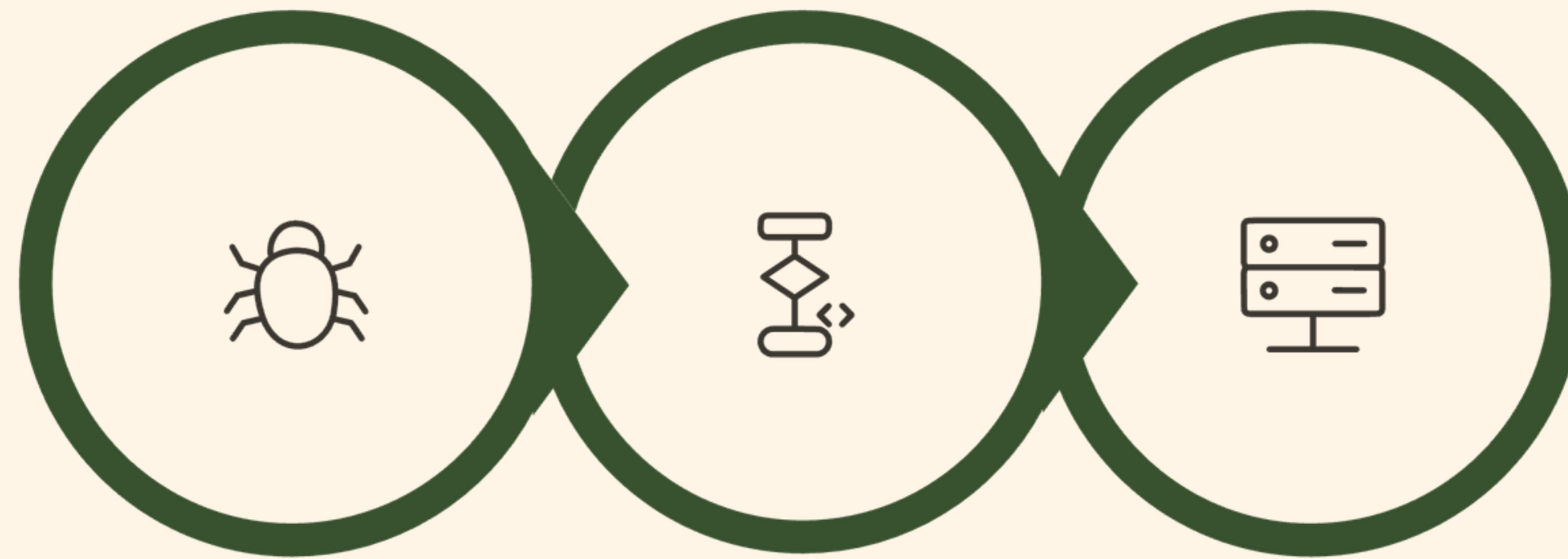
22,000 simultaneous users crashed the system.
Teachers faced 5-minute wait times. Vendors pointed fingers.

407,000
Students tracked

665
Schools connected

\$15M
Invested before failure

Chicago Public Schools: The Architecture Fix



Identify

Redesign

Scale

Scalability must be designed for peak demand — not average usage.

Flexcar: Solving the Bottleneck

1

Manual Chaos

Phone reservations, busy lines, 5-min waits

2

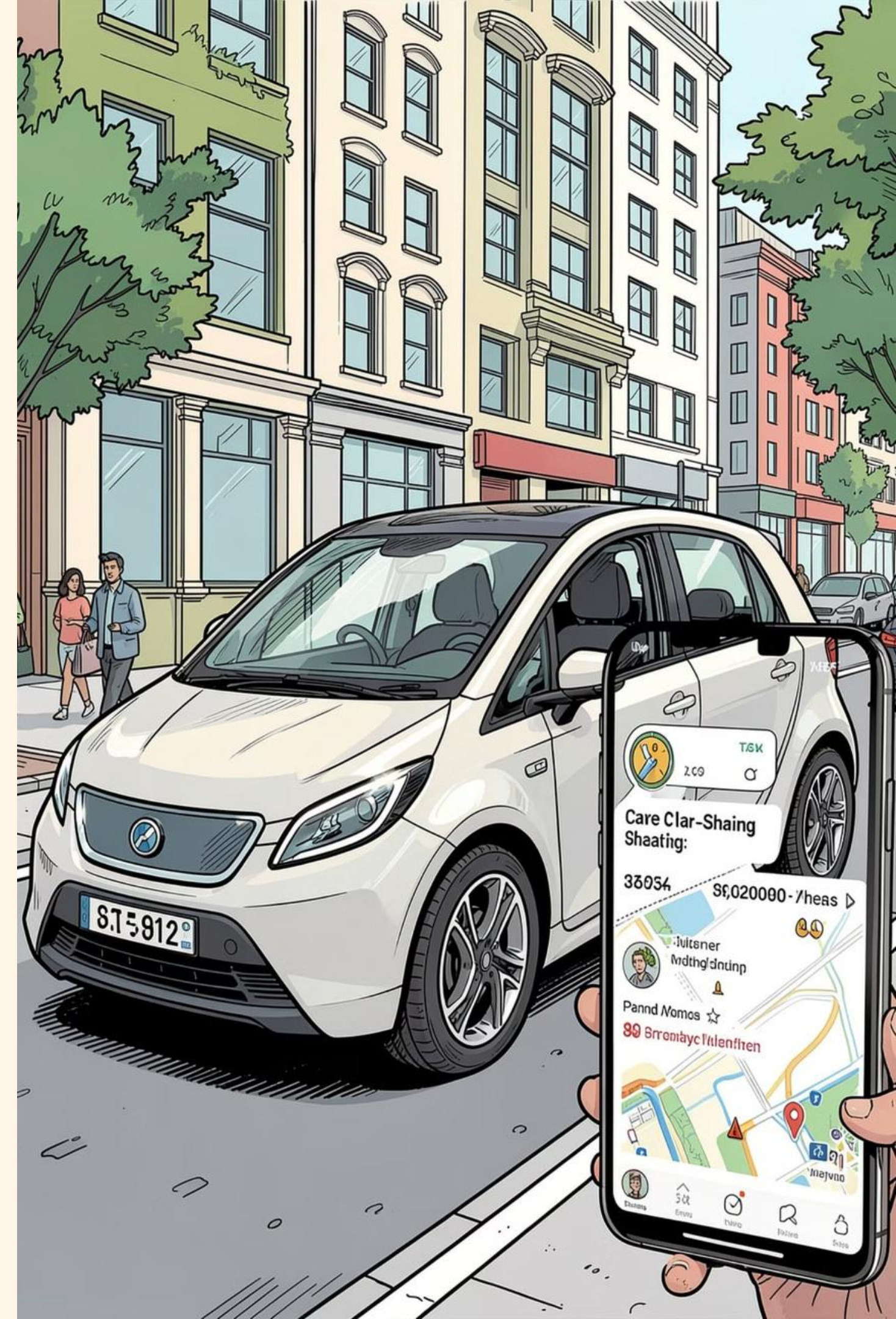
The Gap

Billing errors, inventory miscommunication

3

Digital Shift

Internet reservations, remote entry



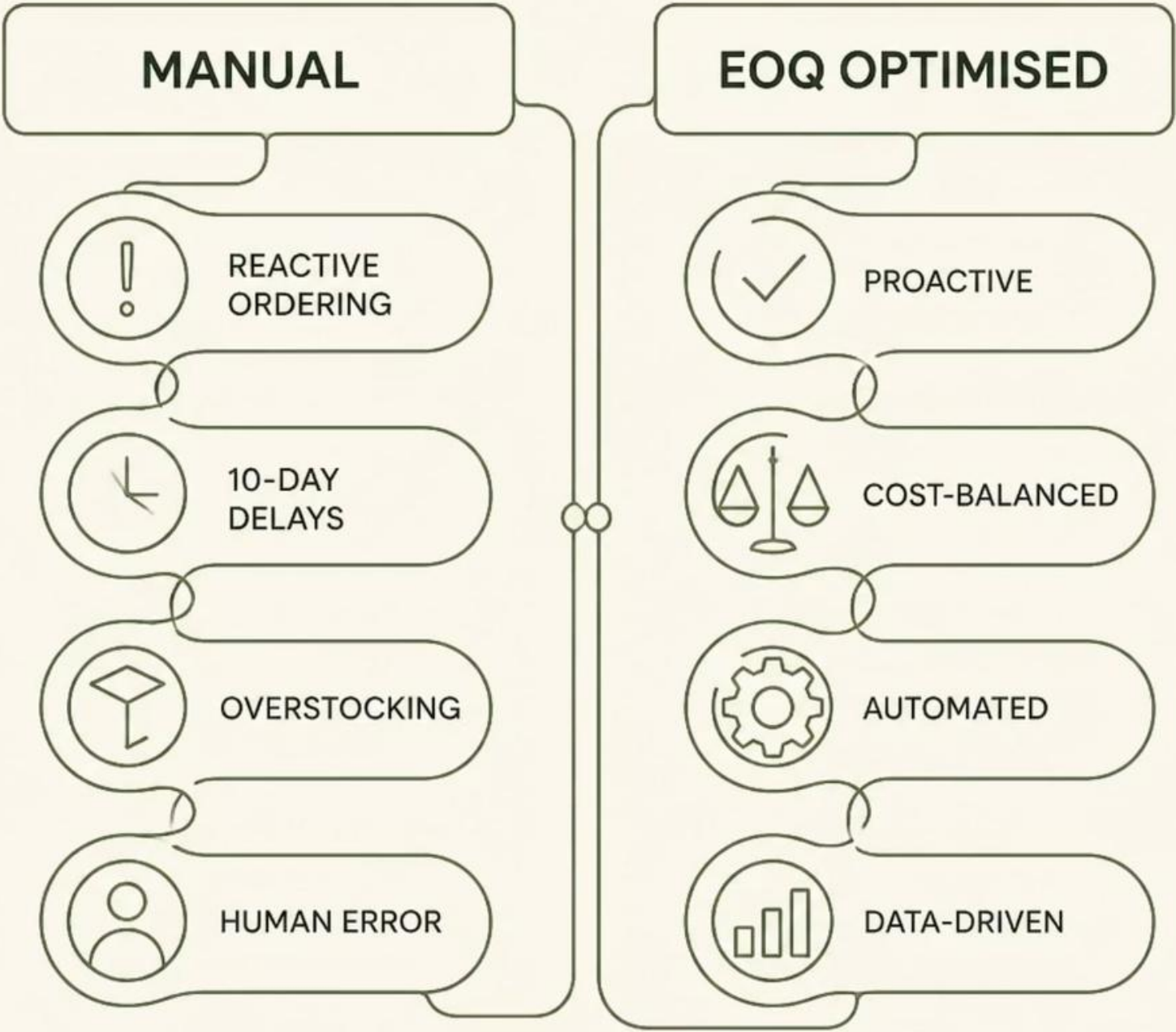
EZPC: Data vs. Knowledge

The Problem

Purchasing manager Stella over-ordered low-demand parts. Manual calculations caused **10-day delays**.

The Solution

Economic Order Quantity (EOQ) models — balancing holding and ordering costs automatically.

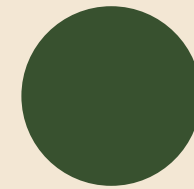


The Business Case Framework



Operational

Will users adopt it?



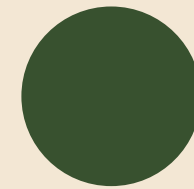
Technical

Can it be built?



Economic

Does ROI justify TCO?



Schedule

Is the timeline realistic?



TCO Trap: Never overlook ongoing maintenance costs when calculating ROI.

Cowboy Entrepreneurship in IT

The Scattergun Trap

Too many unrelated ideas → chaos, cash-flow crises

Heterogeneous Teams Win

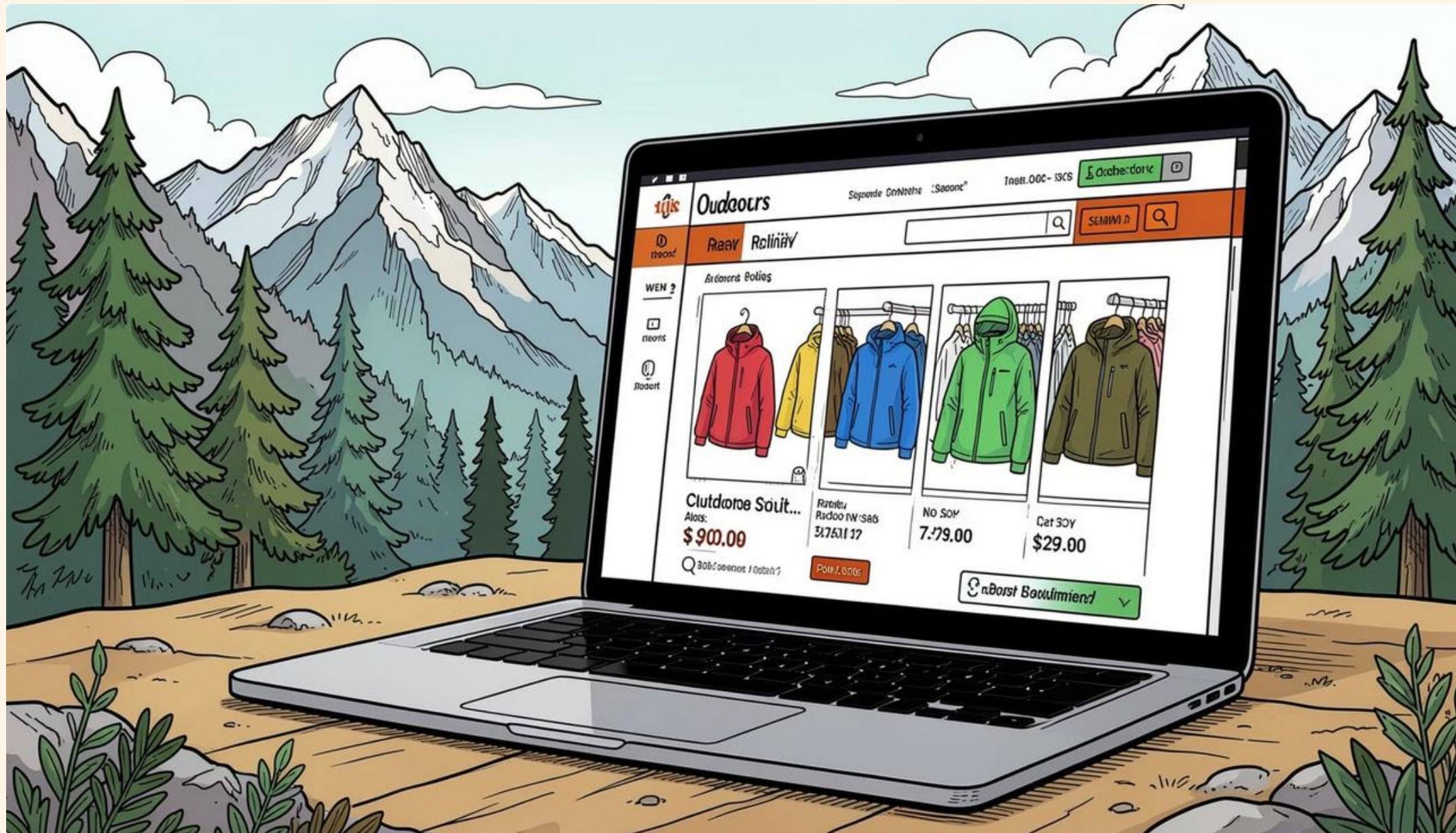
Diverse skills outperform solo founders

The Pivot Point

Selling out demands a management style shift



Strategic Alignment: The RMO Example



Rocky Mountain Outfitters

E-commerce growth aligned with a dedicated customer support system.

01

SWOT Analysis

Validate before writing code

02

IT Serves the Mission

Not the reverse

The Path Forward

Code is not the answer

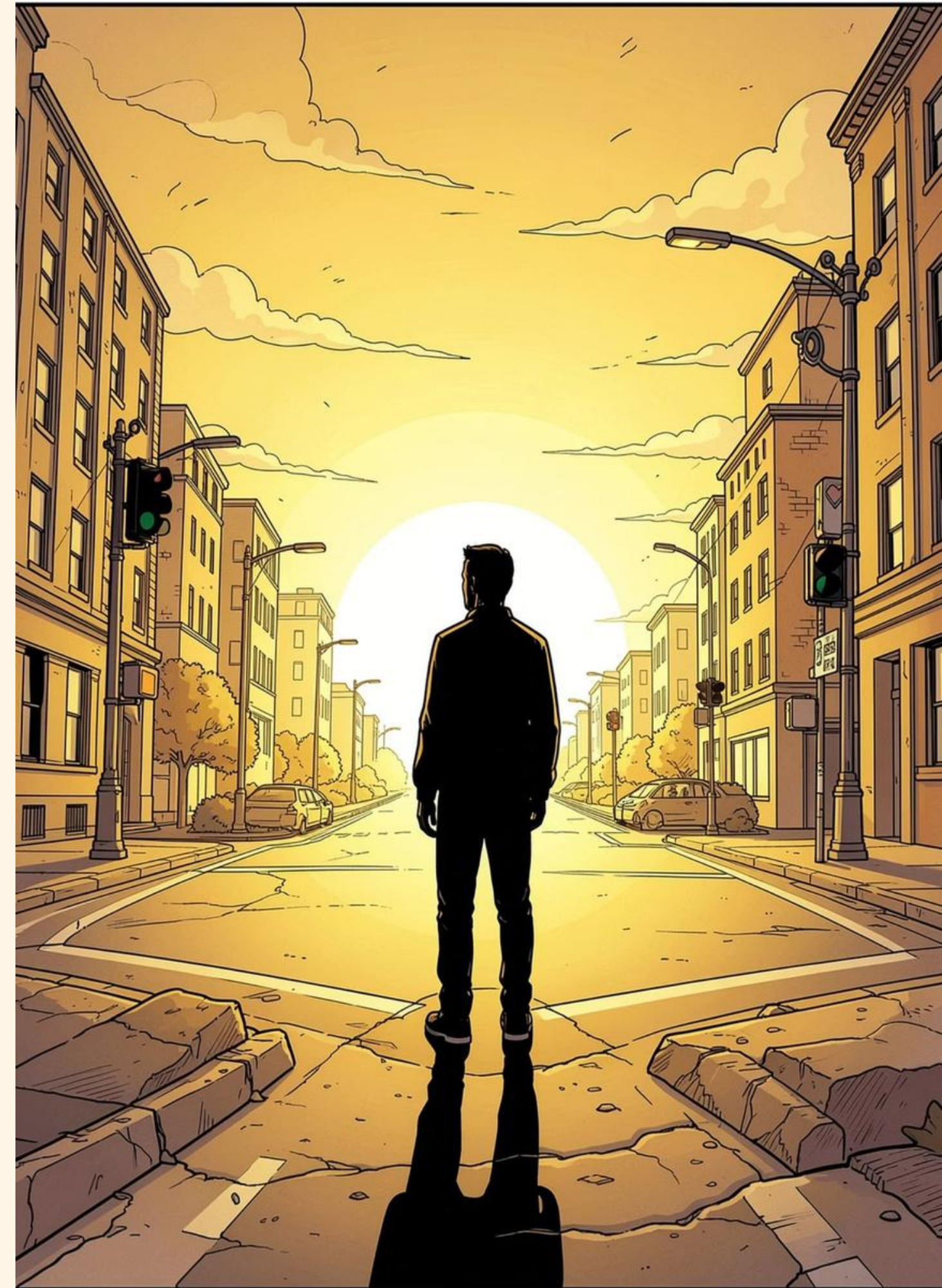
Business case, stakeholders, architecture come first

Build last, ask first

Lead with user requirements

Audit now

Validate feasibility before investing

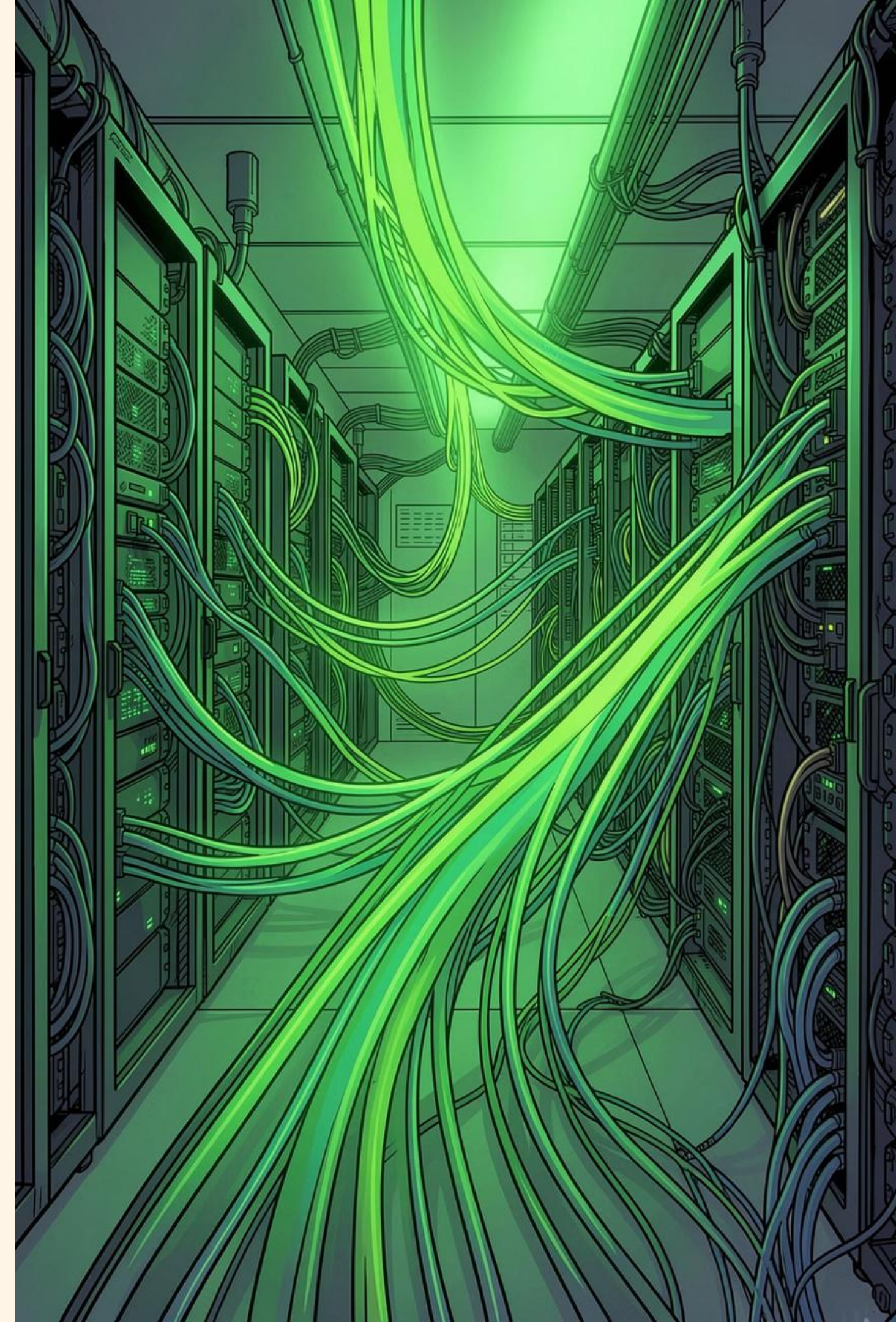


Data Governance & Data Ethics in IT

Foundations for responsible, trustworthy, and compliant data management in the digital age.

IT FUNDAMENTALS

DATA MANAGEMENT



What is Data Governance?

Definition

The framework of policies, standards, and processes that ensure data is managed as a strategic asset.

Core Goal

Ensure data is accurate, accessible, secure, and used responsibly across the organisation.

Why It Matters

Poor governance leads to breaches, compliance failures, and loss of stakeholder trust.



Policies & Procedures



What policies cover

Data classification — sensitivity levels and handling rules

Access controls — who can view, edit, or share data

Retention schedules — how long data is kept and when it's deleted

Incident response — steps to take when a breach occurs

Roles & Responsibilities

Data Owner

Accountable for data quality, classification, and access decisions within a domain.

Data Steward

Ensures day-to-day data quality, metadata accuracy, and policy compliance.

Data Custodian

Manages technical infrastructure — storage, backups, and security controls.

Data Governance Council

Cross-functional body setting strategy, resolving disputes, and overseeing compliance.

Principles of Data Ethics

Fairness

Data systems must not discriminate or produce biased outcomes.

Beneficence

Use data in ways that benefit individuals and society.

Respect

Honour individual autonomy and the right to control personal data.

Integrity

Maintain honesty and consistency in all data practices.



Privacy & Confidentiality

Key distinctions

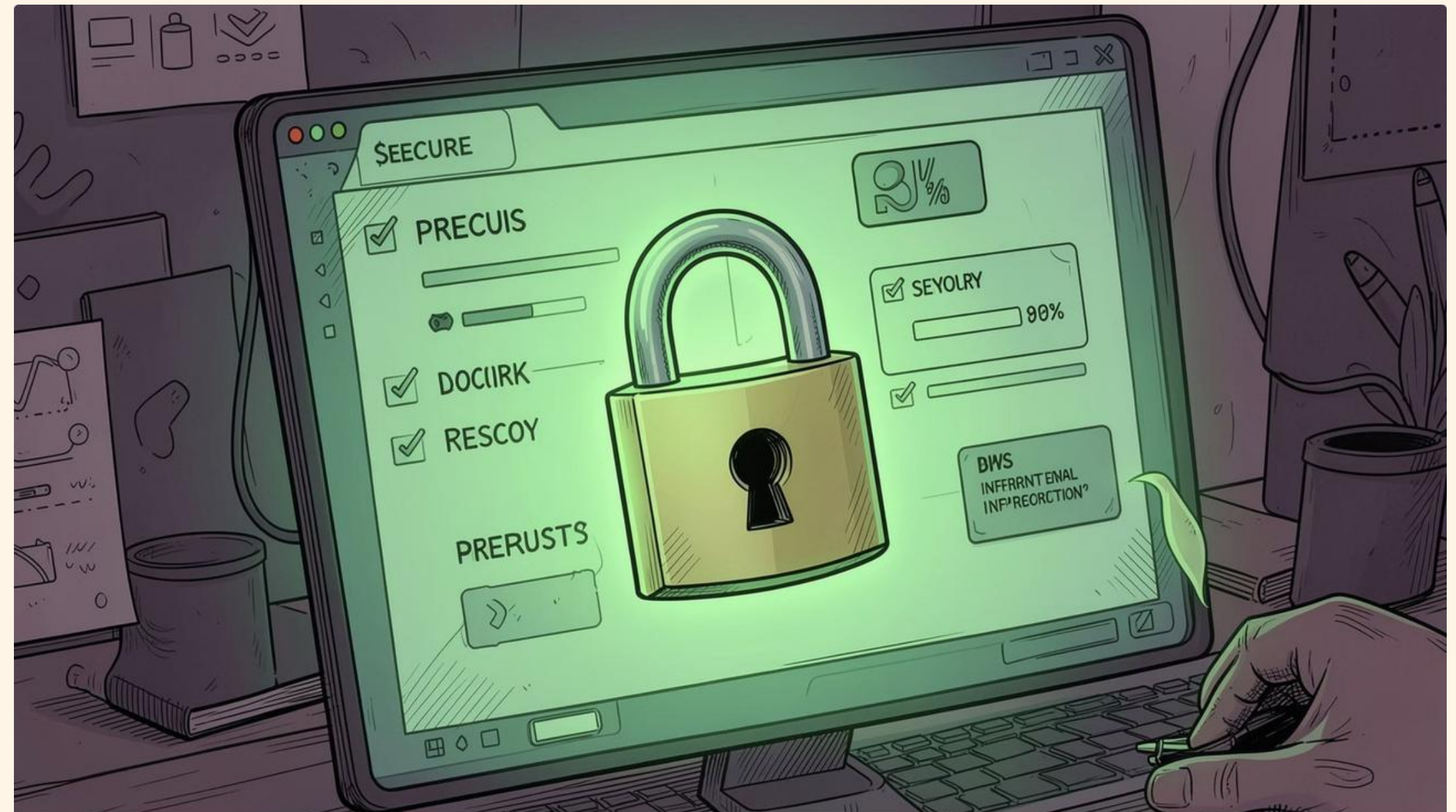
Privacy

An individual's right to control how their personal data is collected and used.

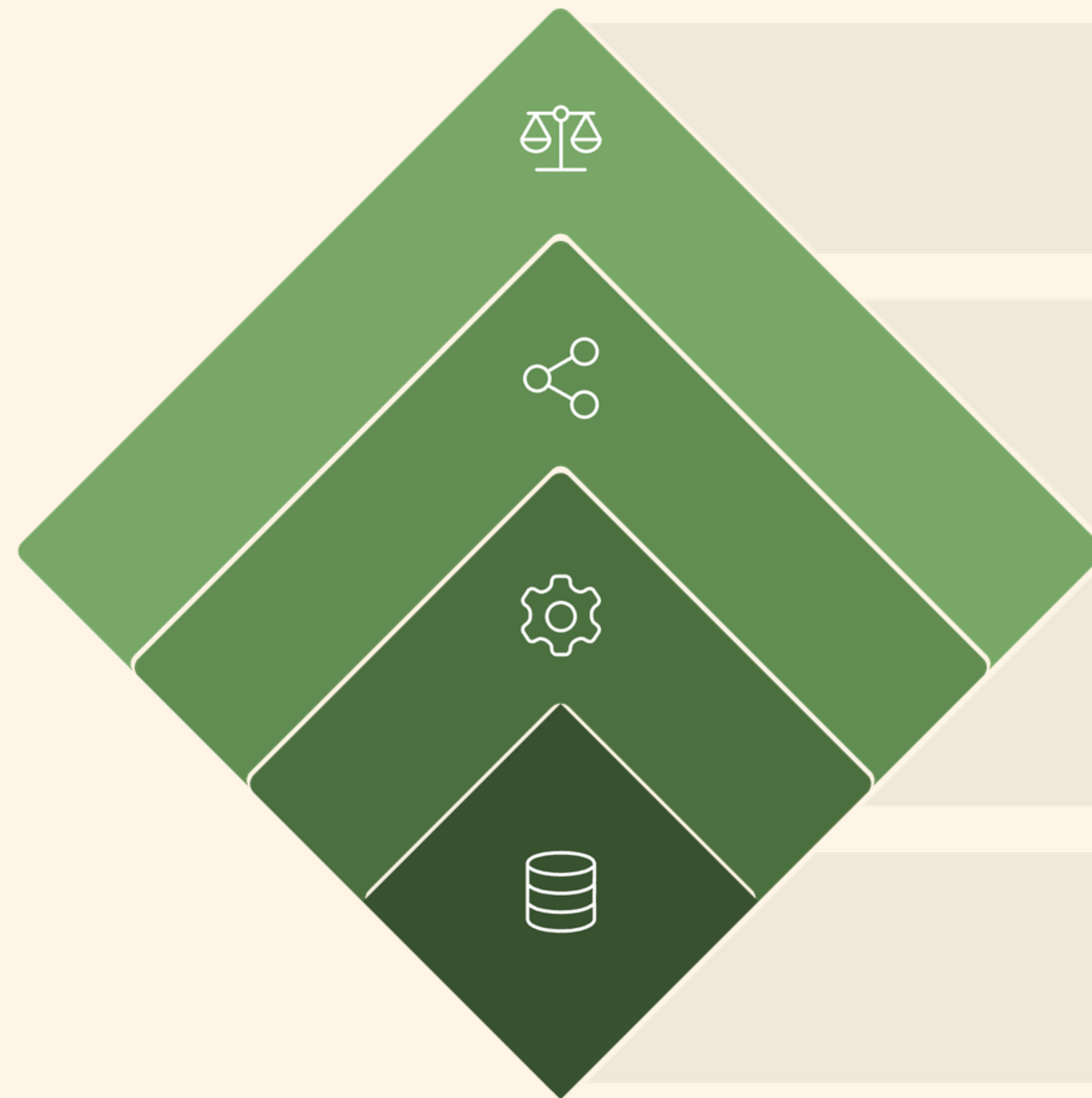
Confidentiality

The obligation to protect data from unauthorised access or disclosure.

- Always collect the **minimum data necessary** — and only with informed consent.



Transparency & Accountability



Accept Responsibility

Be answerable for outcomes

Share Practices

Disclose sharing and access

Clear Usage

Describe how data is used

Open Collection

Explain what data is gathered

Organisations must be able to explain *why* data decisions were made — and accept consequences when things go wrong.

Regulatory Compliance



Why compliance matters

- Legal obligation — non-compliance means heavy fines
- Builds customer trust and brand reputation
- Drives better internal data practices

 Other frameworks include HIPAA (healthcare), PIPEDA (Canada), and the UK Data Protection Act 2018.

Building a Culture of Responsible Data Use

01

Leadership commitment

Executives must champion data ethics from the top down.

02

Training & awareness

Regular, role-specific training for all staff handling data.

03

Embed in workflows

Integrate governance checks into every project and system lifecycle.

04

Measure & improve

Track metrics, audit regularly, and continuously refine practices.

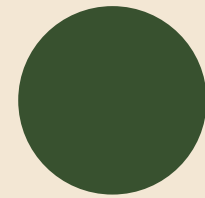


Key Takeaways



Governance is a framework

Policies, roles, and processes working together.



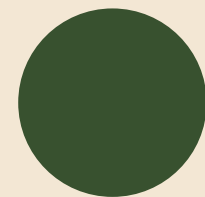
Ethics is non-negotiable

Fairness, privacy, and transparency are core obligations.



Compliance protects everyone

GDPR, CCPA, and similar laws set the baseline.



Culture drives behaviour

Responsible data use must be lived, not just documented.